

Feasible characters: updated tables

Recomputed with `feasgen.g` (FeasChar repository)

Table numbers and captions are those of Chapter 6 of A. J. Litterick, *On non-generic finite subgroups of exceptional algebraic groups*, Mem. Amer. Math. Soc. 1207 (2018). The rows of each table are the feasible characters compatible with the adjoint module, all end-node Weyl modules and, in characteristic p , the irreducible modules $L(\lambda)$ that are smaller than the corresponding Weyl modules. All element orders are used, and the class assignment must be compatible with every power map. Only $L(G)$ and $V_{\min}$ are shown, as in the Memoir.

Rows are listed up to automorphisms of the character table of H (and, for F_4 with $p = 2$, up to the graph automorphism of F_4). Columns are labelled by degree, with letters in the order of the irreducible (Brauer) characters in GAP's character table library. These labels need not agree with the Memoir's.

Provenance. These tables were computed and typeset by code written by an AI system (Claude, Anthropic), October 2026, under the direction of the Memoir's author. They are computational results and have not been independently checked by hand. See the README.

P, **N**: as in the Memoir, carried over from the Memoir row with the same composition factors. **new**: not in the Memoir. Differences from the Memoir are listed in DIFFERENCES.md.

F_4 : feasible characters on $L(G) = V_{52}$ and $V_{\min} = V_{26}$

Table 6.1: $\text{Alt}_{10} < F_4, p = 2$

	V_{52}				V_{26}		
	1	8	16	26	1	8	16
1)	2	1	1	1	2	1	1

Modules tested: V52, L26, V26.

Table 6.2: $\text{Alt}_9 < F_4, p = 2$

	V_{52}					V_{26}			
	1	8_a	8_b	8_c	26	1	8_a	8_b	8_c
1)	2	1	1	1	1	2	1	1	1

Modules tested: V52, L26, V26.

Table 6.3: $\text{Alt}_7 < F_4, p = 5$

	V_{52}			V_{26}	
	8	10_a	10_b	1	8
1)	4	1	1	2	3

Modules tested: V52, V26.

Table 6.4: $\text{Alt}_7 < F_4, p = 2$

	V_{52}						V_{26}			
	1	4_a	4_b	6	14	20	1	4_a	4_b	6
1)	4	2	2	2	0	1	4	2	2	1
2)	4	2	2	3	1	0	4	2	2	1

Memoir rows not reproduced: 5) (not compatible with L26).

Modules tested: V52, L26, V26.

Table 6.5: $\text{Alt}_6 < F_4$, $p = 0$ or $p \nmid |H|$

		V_{52}							V_{26}						
		1	5_a	5_b	8_a	8_b	9	10	1	5_a	5_b	8_a	8_b	9	10
1)	P	0	0	0	4	0	0	2	2	0	0	3	0	0	0
2)		0	0	0	3	1	0	2	0	0	0	0	1	2	0
3)		0	0	0	2	2	0	2	1	0	0	1	1	1	0
4)		2	1	1	0	0	0	4	1	0	0	1	1	1	0
5)		3	0	0	0	0	1	4	0	1	1	1	1	0	0
6)		3	0	0	0	0	1	4	1	3	0	0	0	0	1

Modules tested: V52, V26.

Table 6.6: $\text{Alt}_6 < F_4$, $p = 5$

		V_{52}					V_{26}				
		1	5_a	5_b	8	10	1	5_a	5_b	8	10
1)	P	0	0	0	4	2	2	0	0	3	0
2)		2	1	1	0	4	2	0	0	3	0
3)		4	0	0	1	4	0	1	1	2	0
4)		4	0	0	1	4	1	3	0	0	1

Modules tested: V52, V26.

Table 6.7: $\text{Alt}_5 < F_4$, $p = 0$ or $p \nmid |H|$

		V_{52}					V_{26}				
		1	3_a	3_b	4	5	1	3_a	3_b	4	5
1)	P	0	5	3	2	4	0	0	1	2	3
2)		3	5	3	5	1	0	0	1	2	3
3)		14	1	0	0	7	0	7	0	0	1
4)		0	4	4	2	4	1	1	1	1	3
5)		3	4	4	5	1	1	1	1	1	3
6)		0	6	2	2	4	2	3	0	0	3
7)		3	6	2	5	1	2	3	0	0	3
8)		3	5	3	5	1	3	0	1	5	0
9)		3	4	4	5	1	4	1	1	4	0
10)		3	6	2	5	1	5	3	0	3	0
11)		8	13	0	0	1	8	6	0	0	0

Modules tested: V52, V26.

Table 6.8: $\text{Alt}_5 < F_4$, $p = 3$

		V_{52}				V_{26}			
		1	3_a	3_b	4	1	3_a	3_b	4
1)	N	21	1	0	7	1	7	0	1
2)	P, N	4	5	3	6	3	0	1	5
3)	P, N	4	4	4	6	4	1	1	4
4)	N	4	6	2	6	5	3	0	3
5)		9	13	0	1	8	6	0	0

Modules tested: V52, V26, L25.

Table 6.9: $M_{11} < F_4$, $p = 11$

		V_{52}			V_{26}		
		10_a	10_b	16	1	9	16
1)		1	1	2	1	1	1

Modules tested: V52, V26.

Table 6.10: $J_1 < F_4$, $p = 11$

		V_{52}			V_{26}	
		1	7	14	1	7
1)		3	5	1	5	3

Modules tested: V52, V26.

Table 6.11: $J_2 < F_4$, $p = 2$

		V_{52}					V_{26}	
		1	6_a	6_b	14_a	14_b	1	6_a
1)	N	8	3	2	0	1	8	3
2)	N	8	5	0	1	0	8	3

Modules tested: V52, L26, V26.

Table 6.12: $L_2(7) < F_4$, $p = 0$ or $p \nmid |H|$

		V_{52}						V_{26}					
		1	3_a	3_b	6	7	8	1	3_a	3_b	6	7	8
1)	P	0	1	1	0	2	4	0	1	1	2	0	1
2)		0	1	1	0	2	4	2	0	0	0	0	3
3)		3	1	1	0	5	1	0	1	1	2	0	1
4)		3	1	1	0	5	1	2	0	0	0	0	3
5)		3	1	1	0	5	1	5	0	0	0	3	0
6)		8	0	0	6	0	1	0	3	3	0	0	1
7)		8	6	6	0	0	1	8	3	3	0	0	0

Modules tested: V52, V26.

Table 6.13: $L_2(7) < F_4$, $p = 3$

		V_{52}					V_{26}				
		1	3_a	3_b	6	7	1	3_a	3_b	6	7
1)	P, N	4	1	1	0	6	1	1	1	2	1
2)	N	4	1	1	0	6	5	0	0	0	3
3)		9	0	0	6	1	1	3	3	0	1
4)		9	6	6	0	1	8	3	3	0	0

Memoir rows not reproduced: 3) (not compatible with L25).

Modules tested: V52, V26, L25.

Table 6.14: $L_2(8) < F_4$, $p = 0$ or $p \nmid |H|$

		V_{52}							V_{26}					
		1	7_a	7_b	7_c	7_d	8	9_a	1	7_a	7_b	8	9_a	9_c
1)	P	0	2	1	1	1	1	1	0	0	0	1	1	1
2)		0	2	1	1	1	1	1	1	1	0	0	1	1
3)		1	2	1	1	1	2	0	2	0	0	3	0	0
4)		1	2	1	1	1	2	0	3	1	0	2	0	0
5)		3	1	5	1	0	0	0	5	0	3	0	0	0

Modules tested: V52, V26.

Table 6.15: $L_2(8) < F_4, p = 7$

		V_{52}						V_{26}			
		1	7_a	7_b	7_c	7_d	8	1	7_a	7_b	8
1)	P, N	1	2	1	1	1	2	2	0	0	3
2)	N	1	2	1	1	1	2	3	1	0	2
3)		3	1	5	1	0	0	5	0	3	0

Modules tested: V52, V26.

Table 6.16: $L_2(8) < F_4, p = 3$

		V_{52}			V_{26}			
		1	7	9_a	1	7	9_a	9_c
1)	P, N	1	6	1	1	1	1	1
2)	N	3	7	0	5	3	0	0

Modules tested: V52, V26, L25.

Table 6.17: $L_2(13) < F_4, p = 0$ or $p \nmid |H|$

		V_{52}					V_{26}			
		1	7_a	12_a	12_b	14_b	1	7_a	12_a	14_a
1)	P	0	0	1	1	2	0	0	1	1
2)		3	5	0	0	1	5	3	0	0

Modules tested: V52, V26.

Table 6.18: $L_2(13) < F_4, p = 7$

		V_{52}				V_{26}			
		1	7_a	12	14_b	1	7_a	12	14_a
1)	P	0	0	2	2	0	0	1	1
2)		3	5	0	1	5	3	0	0

Modules tested: V52, V26.

Table 6.19: $L_2(13) < F_4, p = 3$

		V_{52}					V_{26}			
		1	7_a	7_b	12_a	12_b	1	7_a	12_a	13
1)	P	0	2	2	1	1	1	0	1	1
2)		3	6	1	0	0	5	3	0	0

Modules tested: V52, V26, L25.

Table 6.20: $L_2(13) < F_4, p = 2$

		V_{52}						V_{26}			
		1	6_a	6_b	12_a	12_b	14	1	6_a	12_a	14
1)	P	0	0	0	1	1	2	0	0	1	1
2)	N	8	5	0	0	0	1	8	3	0	0
3)	N	8	3	2	0	0	1	8	3	0	0

Modules tested: V52, L26, V26.

Table 6.21: $L_2(17) < F_4, p = 0$ or $p \nmid |H|$

		V_{52}			V_{26}			
		16_a	18_b	18_c	1	9_a	16_a	17
1)	P	1	1	1	0	1	0	1
2)		1	1	1	1	1	1	0

Modules tested: V52, V26.

Table 6.22: $L_2(17) < F_4, p = 3$

		V_{52}			V_{26}		
		16	18_b	18_c	1	9_a	16
1)	P	1	1	1	1	1	1

Modules tested: V52, V26, L25.

Table 6.23: $L_2(17) < F_4, p = 2$

	V_{52}					V_{26}		
	1	8_a	8_b	16_a		1	8_a	8_b
1)	4	2	2	1		2	2	1

Modules tested: V52, L26, V26.

Table 6.24: $L_2(25) < F_4, p \neq 2, 3, 5$

	V_{52}		V_{26}
	26_d	26_e	26_a
1) P	1	1	1

Modules tested: V52, V26.

Table 6.25: $L_2(25) < F_4, p = 3$

	V_{52}	V_{26}	
	26	1	25
1) P	2	1	1

Modules tested: V52, V26, L25.

Table 6.26: $L_2(25) < F_4, p = 2$

	V_{52}	V_{26}
	26	26
1) P	2	1

Modules tested: V52, L26, V26.

Table 6.27: $L_2(27) < F_4, p \neq 2, 3, 7$

	V_{52}		V_{26}
	26_a	26_b	26_e
1) P	1	1	1

Modules tested: V52, V26.

Table 6.28: $L_2(27) < F_4, p = 7$

		V_{52}	V_{26}	
		26	13_a	13_b
1)	P	2	1	1

Modules tested: V52, V26.

Table 6.29: $L_2(27) < F_4, p = 2$

		V_{52}		V_{26}
		26_a	26_b	26_a
1)	P	1	1	1

Modules tested: V52, L26, V26.

Table 6.30: $L_3(3) < F_4, p \neq 2, 3$

		V_{52}		V_{26}
		26_b	26_c	26_a
1)	P	1	1	1

Modules tested: V52, V26.

Table 6.31: $L_3(3) < F_4, p = 2$

		V_{52}	V_{26}
		26	26
1)	P	2	1

Modules tested: V52, L26, V26.

Table 6.32: $L_4(3) < F_4, p = 2$

		V_{52}		V_{26}
		26_a	26_b	26_a
1)	P	1	1	1

Modules tested: V52, L26, V26.

Table 6.33: $U_3(3) < F_4$, $p = 0$ or $p \nmid |H|$

	V_{52}						V_{26}			
	1	7_a	7_b	7_c	14	21_a	1	6	7_a	14
1)	3	5	0	0	1	0	5	0	3	0
2)	3	0	2	2	0	1	0	2	0	1

Modules tested: V52, V26.

Table 6.34: $U_3(3) < F_4$, $p = 7$

	V_{52}								V_{26}				
	1	6	7_a	7_b	7_c	14	21_a	26	1	6	7_a	14	26
1) P	0	2	0	0	0	1	0	1	0	0	0	0	1
2)	3	0	0	2	2	0	1	0	0	2	0	1	0
3)	3	0	5	0	0	1	0	0	5	0	3	0	0

Modules tested: V52, V26.

6.35: ${}^3D_4(2) < F_4$, $p \neq 2, 3$ (irreducible on both)

	V_{52}	V_{26}
	52	26
1) P	1	1

Modules tested: V52, V26.

Table 6.36: ${}^3D_4(2) < F_4$, $p = 3$

	V_{52}	V_{26}
	52	1 25
1) P	1	1 1

Modules tested: V52, V26, L25.

E_6 : feasible characters on $L(G) = V_{78}$ and $V_{\min} = V_{27}$

Table 6.37: $\text{Alt}_{12} < E_6$, $p = 2$

	V_{78}				V_{27}		
	1	16_a	16_b	44	1	10	16_a
1)	2	1	1	1	1	1	1

Modules tested: V_{78} , V_{27} .

Table 6.38: $\text{Alt}_{11} < E_6$, $p = 2$

	V_{78}				V_{27}		
	1	16_a	16_b	44	1	10	16_a
1)	2	1	1	1	1	1	1

Modules tested: V_{78} , V_{27} .

Table 6.39: $\text{Alt}_{10} < E_6$, $p = 2$

	V_{78}				V_{27}			
	1	8	16	26	1	8	16	26
1)	2	1	1	2	1	0	0	1
2)	4	2	2	1	3	1	1	0

Modules tested: V_{78} , V_{27} .

Table 6.40: $\text{Alt}_9 < E_6$, $p = 2$

		V_{78}					V_{27}				
		1	8_a	8_b	8_c	26	1	8_a	8_b	8_c	26
1)	P	2	1	1	1	2	1	0	0	0	1
2)		4	2	2	2	1	3	1	1	1	0

Modules tested: V_{78} , V_{27} .

Table 6.41: $\text{Alt}_7 < E_6$, $p = 0$ or $p \nmid |H|$

		V_{78}						V_{27}	
		1	6	10_a	10_b	14_a	15	6	15
1)		3	1	2	2	1	1	2	1

Modules tested: V78, V27.

Table 6.42: $3 \cdot \text{Alt}_7 < E_6$, $p = 0$ or $p \nmid |H|$

		V_{78}					V_{27}	
		1	10_a	10_b	14_a	21_a	6_b	15_b
1)		3	2	2	1	1	2	1

Modules tested: V78, V27.

Table 6.43: $\text{Alt}_7 < E_6$, $p = 7$

		V_{78}				V_{27}		
		1	5	10	14_a	1	5	10
1)	N	4	2	5	1	2	3	1

Modules tested: V78, V27.

Table 6.44: $3 \cdot \text{Alt}_7 < E_6$, $p = 7$

		V_{78}				V_{27}	
		1	10	14_a	21_a	6_a	15_a
1)		3	4	1	1	2	1

Modules tested: V78, V27.

Table 6.45: $\text{Alt}_7 < E_6$, $p = 5$

		V_{78}							V_{27}				
		1	6	8	10_a	10_b	15	35	1	6	8	13	15
1)	P	0	0	1	0	0	0	2	0	1	1	1	0
2)		3	2	1	2	2	1	0	0	2	0	0	1
3)		2	0	7	1	1	0	0	3	0	3	0	0

Modules tested: V78, V27.

Table 6.46: $3 \cdot \text{Alt}_7 < E_6$, $p = 5$

		V_{78}							V_{27}			
		1	6_a	8	10_a	10_b	13	35	3_a	6_b	15_b	21_a
1)	P	0	0	1	0	0	0	2	0	1	0	1
2)		3	1	2	2	2	1	0	0	2	1	0
3)		14	0	8	0	0	0	0	7	1	0	0

Modules tested: V78, V27.

Table 6.47: $\text{Alt}_7 < E_6$, $p = 3$

		V_{78}						V_{27}	
		1	6	10_a	10_b	13	15	6	15
1)	P, N	4	1	2	2	1	1	2	1

Modules tested: V78, L77, V27.

Table 6.48: $\text{Alt}_7 < E_6$, $p = 2$

		V_{78}						V_{27}					
		1	4_a	4_b	6	14	20	1	4_a	4_b	6	14	20
1)	N	4	2	2	3	0	2	1	0	0	1	0	1
2)		4	2	2	4	1	1	1	0	0	1	0	1
3)	N	4	2	2	5	2	0	1	0	0	2	1	0
4)		8	4	4	3	0	1	5	2	2	1	0	0
5)		8	4	4	4	1	0	5	2	2	1	0	0

Modules tested: V78, V27.

Table 6.49: $3 \cdot \text{Alt}_7 < E_6$, $p = 2$

		V_{78}						V_{27}	
		1	4_a	4_b	6_a	14	20	6_b	15_a
1)	N	4	2	2	3	0	2	2	1
2)		4	2	2	4	1	1	2	1

Modules tested: V78, V27.

Table 6.50: $\text{Alt}_6 < E_6$, $p = 0$ or $p \nmid |H|$

		V_{78}							V_{27}						
		1	5_a	5_b	8_a	8_b	9	10	1	5_a	5_b	8_a	8_b	9	10
1)	P	0	0	0	3	2	2	2	0	1	1	0	1	1	0
2)		2	1	1	1	0	2	4	0	1	1	0	1	1	0
3)		0	0	0	3	2	2	2	1	0	0	0	1	2	0
4)		2	1	1	1	0	2	4	1	0	0	0	1	2	0
5)		1	0	0	3	3	1	2	1	1	1	1	1	0	0
6)		3	1	1	1	1	1	4	1	1	1	1	1	0	0
7)		1	0	0	3	3	1	2	2	0	0	1	1	1	0
8)		3	1	1	1	1	1	4	2	0	0	1	1	1	0
9)		4	3	0	0	0	1	5	2	3	0	0	0	0	1
10)		2	0	0	7	0	0	2	3	0	0	3	0	0	0

Modules tested: V78, V27.

Table 6.51: $3 \cdot \text{Alt}_6 < E_6$, $p = 0$ or $p \nmid |H|$

		V_{78}							V_{27}				
		1	5_a	5_b	8_a	8_b	9_a	10	3_a	3_c	6_a	9_b	15_a
1)		1	0	0	3	3	1	2	0	0	2	0	1
2)		3	1	1	1	1	1	4	0	0	2	0	1
3)	P	0	0	0	3	2	2	2	0	1	1	2	0
4)		2	1	1	1	0	2	4	0	1	1	2	0
5)		2	0	0	7	0	0	2	3	0	3	0	0
6)		14	0	0	8	0	0	0	7	0	1	0	0
7)		1	0	0	3	3	1	2	1	1	2	1	0
8)		3	1	1	1	1	1	4	1	1	2	1	0
9)		8	0	0	1	1	6	0	3	3	0	1	0

Modules tested: V78, V27.

Table 6.52: $\text{Alt}_6 < E_6$, $p = 5$

		V_{78}					V_{27}				
		1	5_a	5_b	8	10	1	5_a	5_b	8	10
1)	P, N	2	0	0	7	2	1	1	1	2	0

		V_{78}					V_{27}				
		1	5_a	5_b	8	10	1	5_a	5_b	8	10
2)	N	4	1	1	3	4	1	1	1	2	0
3)		5	3	0	1	5	2	3	0	0	1
4)	N	2	0	0	7	2	3	0	0	3	0
5)	N	4	1	1	3	4	3	0	0	3	0

Modules tested: V_{78} , V_{27} .

Table 6.53: $3 \cdot \text{Alt}_6 < E_6$, $p = 5$

		V_{78}					V_{27}		
		1	5_a	5_b	8	10	3_a	6_a	15_a
1)	P, N	2	0	0	7	2	0	2	1
2)	N	4	1	1	3	4	0	2	1
3)	P, N	2	0	0	7	2	3	3	0
4)	N	4	1	1	3	4	3	3	0
5)	N	14	0	0	8	0	7	1	0

Modules tested: V_{78} , V_{27} .

Table 6.54: $\text{Alt}_5 < E_6$, $p = 0$ or $p \nmid |H|$

		V_{78}					V_{27}				
		1	3_a	3_b	4	5	1	3_a	3_b	4	5
1)		5	1	1	3	11	0	3	3	1	1
2)		8	1	1	6	8	0	3	3	1	1
3)		0	5	4	4	7	1	0	1	2	3
4)		3	5	4	7	4	1	0	1	2	3
5)		14	8	0	0	8	1	7	0	0	1
6)		1	5	5	3	7	2	1	1	1	3
7)		4	5	5	6	4	2	1	1	1	3
8)		2	9	2	2	7	3	3	0	0	3
9)		5	9	2	5	4	3	3	0	0	3
10)		6	5	4	10	1	4	0	1	5	0
11)		7	5	5	9	1	5	1	1	4	0
12)		8	9	2	8	1	6	3	0	3	0
13)		16	19	0	0	1	9	6	0	0	0

Modules tested: V_{78} , V_{27} .

Table 6.55: $\text{Alt}_5 < E_6, p = 3$

		V_{78}				V_{27}			
		1	3_a	3_b	4	1	3_a	3_b	4
1)	P, N	15	1	0	15	0	2	3	3
2)	N	16	1	1	14	1	3	3	2
3)	N	22	8	0	8	2	7	0	1
4)	P, N	7	5	4	11	4	0	1	5
5)	N	8	5	5	10	5	1	1	4
6)	N	9	9	2	9	6	3	0	3
7)		17	19	0	1	9	6	0	0

Modules tested: V78, L77, V27.

Table 6.56: $M_{11} < E_6, p = 0$ or $p \nmid |H|$

		V_{78}				V_{27}			
		1	16_a	16_b	45	1	10_a	11	16_a
1)		1	1	1	1	0	0	1	1
2)		1	1	1	1	1	1	0	1

Modules tested: V78, V27.

Table 6.57: $M_{11} < E_6, p = 11$

		V_{78}					V_{27}			
		1	9	10_a	10_b	16	1	9	11	16
1)		1	1	1	1	3	0	0	1	1
2)		1	1	1	1	3	2	1	0	1

Modules tested: V78, V27.

Table 6.58: $M_{11} < E_6, p = 5$

		V_{78}				V_{27}			
		1	16_a	16_b	45	1	10_a	11	16_a
1)	P, N	1	1	1	1	0	0	1	1
2)	N	1	1	1	1	1	1	0	1

Modules tested: V78, V27.

Table 6.59: $M_{11} < E_6$, $p = 3$

		V_{78}							V_{27}					
		1	5_a	5_b	10_b	10_c	24	45	1	5_a	5_b	10_a	10_b	10_c
1)	P, N	3	1	1	1	1	0	1	2	1	0	1	0	1
2)	P, N	3	1	1	1	1	0	1	2	2	1	0	0	1
3)	P, N	4	1	1	2	2	1	0	2	2	1	0	1	0

Modules tested: V78, L77, V27.

Table 6.60: $M_{11} < E_6$, $p = 2$

		V_{78}					V_{27}		
		1	10	16_a	16_b	44	1	10	16_a
1)		2	0	1	1	1	1	1	1
2)	N	4	3	0	0	1	1	1	1

Modules tested: V78, V27.

Table 6.61: $M_{12} < E_6$, $p = 5$

		V_{78}	V_{27}	
		78	11_a	16_a
1)	P	1	1	1

Modules tested: V78, V27.

Table 6.62: $M_{12} < E_6$, $p = 2$

		V_{78}				V_{27}		
		1	16_a	16_b	44	1	10	16_a
1)		2	1	1	1	1	1	1

Modules tested: V78, V27.

Table 6.63: $3 \cdot M_{22} < E_6$, $p = 2$

		V_{78}				V_{27}	
		1	10_a	10_b	34	6_a	15_a
1)	N	4	2	2	1	2	1

Modules tested: V78, V27.

Table 6.64: $J_1 < E_6$, $p = 11$

		V_{78}				V_{27}		
		1	7	14	64	1	7	27
1)	P	0	0	1	1	0	0	1
2)		8	8	1	0	6	3	0

Modules tested: V78, V27.

Table 6.65: $J_2 < E_6$, $p = 2$

		V_{78}					V_{27}			
		1	6_a	6_b	14_a	64_b	1	6_a	6_b	14_a
1)	N	8	4	3	2	0	1	2	0	1
2)	P	0	0	0	1	1	1	1	1	1
3)	N	16	8	0	1	0	9	3	0	0

Modules tested: V78, V27.

Table 6.66: $3 \cdot J_3 < E_6$, $p = 2$

		V_{78}	V_{27}	
		78_a	9_a	18_c
1)	P	1	1	1

Modules tested: V78, V27.

6.67: $3 \cdot Fi_{22} < E_6$, $p = 2$ (irreducible on both)

	V_{78} 78	V_{27} 27_a
1) P	1	1

Modules tested: V78, V27.

Table 6.68: $L_2(7) < E_6$, $p = 0$ or $p \nmid |H|$

		V_{78}						V_{27}					
		1	3_a	3_b	6	7	8	1	3_a	3_b	6	7	8
1)	P	0	2	2	2	2	5	0	0	0	2	1	1
2)		0	2	2	2	2	5	1	1	1	2	0	1
3)		2	1	1	0	2	7	3	0	0	0	0	3
4)		3	2	2	2	5	2	0	0	0	2	1	1
5)		3	2	2	2	5	2	1	1	1	2	0	1
6)		5	1	1	0	5	4	3	0	0	0	0	3
7)		8	1	1	0	8	1	6	0	0	0	3	0
8)		8	3	3	6	0	2	1	3	3	0	0	1
9)		14	0	0	0	0	8	0	7	0	1	0	0
10)		16	9	9	0	0	1	9	3	3	0	0	0

Modules tested: V78, V27.

Table 6.69: $L_2(7) < E_6$, $p = 3$

		V_{78}					V_{27}				
		1	3_a	3_b	6	7	1	3_a	3_b	6	7
1)	P, N	5	2	2	2	7	1	0	0	2	2
2)	N	5	2	2	2	7	2	1	1	2	1
3)	N	9	1	1	0	9	6	0	0	0	3
4)	N	10	3	3	6	2	2	3	3	0	1
5)		17	9	9	0	1	9	3	3	0	0
6)	N	22	0	0	0	8	0	7	0	1	0

Modules tested: V78, L77, V27.

Table 6.70: $L_2(8) < E_6$, $p = 0$ or $p \nmid |H|$

		V_{78}									V_{27}						
		1	7_a	7_b	7_c	7_d	8	9_a	9_b	9_c	1	7_a	7_b	8	9_a	9_b	9_c
1)	P	0	2	1	1	1	2	2	0	1	1	0	0	1	1	0	1
2)		0	2	1	1	1	2	1	1	1	0	0	0	0	1	1	1
3)		1	3	1	1	1	1	2	0	1	2	1	0	0	1	0	1
4)		3	2	1	1	1	5	0	0	0	3	0	0	3	0	0	0
5)		4	3	1	1	1	4	0	0	0	4	1	0	2	0	0	0
6)		8	1	8	1	0	0	0	0	0	6	0	3	0	0	0	0

Modules tested: V78, V27.

Table 6.71: $L_2(8) < E_6$, $p = 7$

		V_{78}						V_{27}			
		1	7_a	7_b	7_c	7_d	8	1	7_a	7_b	8
1)	N	3	2	1	1	1	5	3	0	0	3
2)	N	4	3	1	1	1	4	4	1	0	2
3)		8	1	8	1	0	0	6	0	3	0

Modules tested: V78, V27.

Table 6.72: $L_2(8) < E_6$, $p = 3$

		V_{78}					V_{27}				
		1	7	9_a	9_b	9_c	1	7	9_a	9_b	9_c
1)	N	2	7	2	0	1	2	1	1	0	1
2)	P, N	2	7	1	1	1	0	0	1	1	1
3)	N	8	10	0	0	0	6	3	0	0	0

Modules tested: V78, L77, V27.

Table 6.73: $L_2(11) < E_6$, $p = 0$ or $p \nmid |H|$

		V_{78}							V_{27}						
		1	5_a	5_b	10_a	11	12_a	12_b	1	5_a	5_b	10_a	10_b	11	12_b
1)	P	0	1	1	1	2	2	1	0	1	0	0	1	0	1
2)		1	1	1	1	3	1	1	1	1	0	0	1	1	0
3)		4	1	1	4	0	1	1	2	2	1	1	0	0	0

Modules tested: V78, V27.

Table 6.74: $L_2(11) < E_6, p = 5$

		V_{78}					V_{27}					
		1	5_a	5_b	10_a	11	1	5_a	5_b	10_a	10_b	11
1)	N	3	1	1	1	5	1	1	0	0	1	1
2)	N	6	1	1	4	2	2	2	1	1	0	0

Modules tested: V78, V27.

Table 6.75: $L_2(11) < E_6, p = 3$

		V_{78}						V_{27}				
		1	5_a	5_b	10	12_a	12_b	1	5_a	5_b	10	12_b
1)	P, N	2	1	1	3	2	1	0	2	1	0	1
2)	N	4	1	1	4	1	1	2	2	1	1	0

Modules tested: V78, L77, V27.

Table 6.76: $L_2(11) < E_6, p = 2$

		V_{78}						V_{27}				
		1	5_a	5_b	10	12_a	12_b	1	5_a	5_b	10	12_b
1)		2	0	0	4	2	1	0	1	0	1	1
2)	P, N	2	3	3	1	2	1	0	1	0	1	1
3)	N	4	1	1	4	1	1	2	2	1	1	0
4)	P, N	4	4	4	1	1	1	2	2	1	1	0

Modules tested: V78, V27.

Table 6.77: $L_2(13) < E_6, p = 0$ or $p \nmid |H|$

		V_{78}									V_{27}				
		1	7_a	7_b	12_a	12_b	12_c	13	14_a	14_b	1	7_a	12_a	13	14_a
1)	P	0	0	0	1	1	1	0	1	2	0	0	0	1	1
2)		0	0	0	2	1	0	0	1	2	1	0	1	0	1
3)		1	2	2	1	1	1	1	0	0	0	0	0	1	1
4)		1	2	2	2	1	0	1	0	0	1	0	1	0	1

	V_{78}									V_{27}				
	1	7_a	7_b	12_a	12_b	12_c	13	14_a	14_b	1	7_a	12_a	13	14_a
5)	8	8	0	0	0	0	0	0	1	6	3	0	0	0

Modules tested: V78, V27.

Table 6.78: $L_2(13) < E_6$, $p = 7$

		V_{78}						V_{27}			
		1	7_a	7_b	12	14_a	14_b	1	7_a	12	14_a
1)		0	0	0	3	1	2	1	0	1	1
2)	N	2	2	2	4	0	0	1	0	1	1
3)	N	8	8	0	0	0	1	6	3	0	0

Modules tested: V78, V27.

Table 6.79: $L_2(13) < E_6$, $p = 3$

		V_{78}							V_{27}			
		1	7_a	7_b	12_a	12_b	12_c	13	1	7_a	12_a	13
1)	P	1	2	2	1	1	1	1	1	0	0	2
2)		1	2	2	2	1	0	1	2	0	1	1
3)		8	9	1	0	0	0	0	6	3	0	0

Modules tested: V78, L77, V27.

Table 6.80: $L_2(13) < E_6$, $p = 2$

		V_{78}							V_{27}				
		1	6_a	6_b	12_a	12_b	12_c	14	1	6_a	6_b	12_a	14
1)	P	0	0	0	1	1	1	3	1	1	1	0	1
2)		0	0	0	2	1	0	3	1	0	0	1	1
3)		2	1	0	0	0	0	5	1	2	0	0	1
4)	N	6	3	3	1	1	1	0	1	1	1	0	1
5)	N	6	3	3	2	1	0	0	1	0	0	1	1
6)	N	8	4	3	0	0	0	2	1	2	0	0	1
7)	N	16	8	0	0	0	0	1	9	3	0	0	0

Modules tested: V78, V27.

Table 6.81: $L_2(17) < E_6$, $p = 0$ or $p \nmid |H|$

		V_{78}						V_{27}				
		1	9_a	16_a	17	18_b	18_c	1	9_a	16_a	17	18_a
1)	P	0	1	1	1	1	1	0	1	0	0	1
2)		0	1	1	1	1	1	1	1	0	1	0
3)		1	1	2	0	1	1	2	1	1	0	0

Modules tested: V78, V27.

Table 6.82: $L_2(17) < E_6$, $p = 3$

		V_{78}					V_{27}			
		1	9_a	16	18_b	18_c	1	9_a	16	18_a
1)	P, N	1	1	2	1	1	0	1	0	1
2)	N	1	1	2	1	1	2	1	1	0

Modules tested: V78, L77, V27.

Table 6.83: $L_2(17) < E_6$, $p = 2$

		V_{78}				V_{27}			
		1	8_a	8_b	16_a	1	8_a	8_b	16_a
1)	N	6	3	2	2	3	1	0	1
2)	N	6	4	3	1	3	2	1	0

Modules tested: V78, V27.

Table 6.84: $L_2(19) < E_6$, $p = 0$ or $p \nmid |H|$

		V_{78}				V_{27}	
		18_a	20_b	20_c	20_d	9_a	18_d
1)	P	1	1	1	1	1	1

Modules tested: V78, V27.

Table 6.85: $L_2(19) < E_6$, $p = 5$

		V_{78}				V_{27}	
		18	20 _b	20 _c	20 _d	9 _a	9 _b
1)	P	1	1	1	1	2	1

Modules tested: V78, V27.

Table 6.86: $L_2(19) < E_6$, $p = 3$

		V_{78}			V_{27}	
		1	18 _a	19	9 _a	18 _d
1)	P	3	1	3	1	1

Modules tested: V78, L77, V27.

Table 6.87: $L_2(19) < E_6$, $p = 2$

		V_{78}				V_{27}	
		18 _a	20 _b	20 _c	20 _d	9 _a	18 _b
1)	P	1	1	1	1	1	1

Modules tested: V78, V27.

Table 6.88: $L_2(25) < E_6$, $p = 0$ or $p \nmid |H|$

		V_{78}						V_{27}		
		1	25	26 _a	26 _b	26 _d	26 _e	1	26 _a	26 _c
1)		0	0	1	0	1	1	1	1	0
2)		1	1	0	2	0	0	1	1	0
3)		1	1	0	2	0	0	1	0	1

Modules tested: V78, V27.

Table 6.89: $L_2(25) < E_6, p = 13$

	V_{78}						V_{27}		
	1	24	26_a	26_b	26_d	26_e	1	26_a	26_c
1)	0	0	1	0	1	1	1	1	0
2)	2	1	0	2	0	0	1	1	0
3)	2	1	0	2	0	0	1	0	1

Modules tested: V78, V27.

Table 6.90: $L_2(25) < E_6, p = 3$

	V_{78}			V_{27}			
	1	25	26	1	13_a	13_b	25
1)	1	1	2	1	1	1	0
2)	1	1	2	2	0	0	1

Modules tested: V78, L77, V27.

Table 6.91: $L_2(25) < E_6, p = 2$

	V_{78}				V_{27}	
	1	12_a	12_b	26	1	26
1)	0	0	0	3	1	1
2)	6	3	3	0	1	1

Modules tested: V78, V27.

Table 6.92: $L_2(27) < E_6, p = 0$ or $p \nmid |H|$

	V_{78}			V_{27}	
	26_a	26_b	26_e	1	26_e
1)	1	1	1	1	1

Modules tested: V78, V27.

Table 6.93: $L_2(27) < E_6$, $p = 13$

	V_{78}			V_{27}	
	26_a	26_b	26_e	1	26_e
1)	1	1	1	1	1

Modules tested: V78, V27.

Table 6.94: $L_2(27) < E_6$, $p = 7$

	V_{78}			V_{27}		
	13_a	13_b	26	1	13_a	13_b
1)	1	1	2	1	1	1

Modules tested: V78, V27.

Table 6.95: $L_2(27) < E_6$, $p = 2$

	V_{78}			V_{27}			
	26_a	26_b	26_c	1	13_a	13_b	26_a
1)	1	1	1	1	1	1	0
2)	2	0	1	1	0	0	1

Modules tested: V78, V27.

Table 6.96: $L_3(3) < E_6$, $p = 0$ or $p \nmid |H|$

		V_{78}									V_{27}		
		1	13	16_a	16_b	16_c	16_d	26_a	26_b	26_c	1	26_a	27
1)	P	0	0	0	0	0	0	1	1	1	0	0	1
2)		0	0	0	0	0	0	1	1	1	1	1	0
3)		1	1	1	1	1	1	0	0	0	0	0	1
4)		1	1	1	1	1	1	0	0	0	1	1	0

Modules tested: V78, V27.

Table 6.97: $L_3(3) < E_6, p = 13$

		V_{78}						V_{27}			
		1	13	16	26_a	26_b	26_c	1	11	16	26_a
1)	P	0	0	0	1	1	1	0	1	1	0
2)		0	0	0	1	1	1	1	0	0	1
3)		1	1	4	0	0	0	0	1	1	0
4)		1	1	4	0	0	0	1	0	0	1

Modules tested: V78, V27.

Table 6.98: $L_3(3) < E_6, p = 2$

		V_{78}							V_{27}	
		1	12	16_a	16_b	16_c	16_d	26	1	26
1)		0	0	0	0	0	0	3	1	1
2)		2	1	1	1	1	1	0	1	1

Modules tested: V78, V27.

Table 6.99: $L_4(3) < E_6, p = 2$

		V_{78}		V_{27}	
		26_a	26_b	1	26_a
1)		2	1	1	1

Modules tested: V78, V27.

Table 6.100: $U_3(3) < E_6, p = 0$ or $p \nmid |H|$

		V_{78}										V_{27}					
		1	6	7_a	7_b	7_c	14	21_a	28_a	28_b	32_a	32_b	1	6	7_a	14	27
1)	P	0	0	0	0	0	1	0	0	0	1	1	0	0	0	0	1
2)		1	0	1	1	1	0	0	1	1	0	0	0	0	0	0	1
3)		3	2	0	2	2	1	1	0	0	0	0	1	2	0	1	0
4)		8	0	8	0	0	1	0	0	0	0	0	6	0	3	0	0

Modules tested: V78, V27.

Table 6.101: $U_3(3) < E_6, p = 7$

	V ₇₈										V ₂₇				
	1	6	7 _a	7 _b	7 _c	14	21 _a	26	28 _a	28 _b	1	6	7 _a	14	26
1)	0	2	0	0	0	1	0	2	0	0	1	0	0	0	1
2)	1	0	1	1	1	0	0	0	1	1	1	0	0	0	1
3)	3	2	0	2	2	1	1	0	0	0	1	2	0	1	0
4)	8	0	8	0	0	1	0	0	0	0	6	0	3	0	0

Modules tested: V78, V27.

Table 6.102: $U_4(2) < E_6, p = 0$ or $p \nmid |H|$

	V ₇₈								V ₂₇					
	1	5 _a	5 _b	10 _a	10 _b	15 _a	20	24	1	5 _a	5 _b	6	10 _b	15 _a
1)	3	0	0	2	2	1	1	0	0	0	0	2	0	1
2)	4	1	1	2	2	0	0	1	2	2	1	0	1	0

Modules tested: V78, V27.

Table 6.103: $U_4(2) < E_6, p = 5$

	V ₇₈								V ₂₇					
	1	5 _a	5 _b	10 _a	10 _b	15 _a	20	23	1	5 _a	5 _b	6	10 _b	15 _a
1)	3	0	0	2	2	1	1	0	0	0	0	2	0	1
2)	5	1	1	2	2	0	0	1	2	2	1	0	1	0

Modules tested: V78, V27.

Table 6.104: $3 \cdot U_4(3) < E_6, p = 2$

	V ₇₈			V ₂₇	
	1	20	34 _a	6 _a	15 _a
1) N	4	2	1	2	1

Modules tested: V78, V27.

.105: $3 \cdot \Omega_7(3) < E_6$, $p = 2$ (irreducible on both)

		V_{78}	V_{27}
		78	27_a
1)	P	1	1

Modules tested: V78, V27.

.106: $3 \cdot G_2(3) < E_6$, $p = 2$ (irreducible on both)

		V_{78}	V_{27}
		78	27_a
1)	P	1	1

Modules tested: V78, V27.

Table 6.107: ${}^3D_4(2) < E_6$, $p \neq 2, 3$

		V_{78}		V_{27}	
		26	52	1	26
1)		1	1	1	1

Modules tested: V78, V27.

Table 6.108: ${}^3D_4(2) < E_6$, $p = 3$

		V_{78}			V_{27}	
		1	25	52	1	25
1)		1	1	1	2	1

Modules tested: V78, L77, V27.

.109: ${}^2F_4(2)' < E_6$, $p \neq 2, 3$ (irreducible on both)

		V_{78}	V_{27}
		78	27_a
1)	P	1	1

Modules tested: V78, V27.

Table 6.110: ${}^2F_4(2)' < E_6, p = 3$

		V_{78}		V_{27}
		1	77	27_a
1)	P	1	1	1

Modules tested: V78, L77, V27.

E_7 : feasible characters on $L(G) = V_{133}$ and $V_{\min} = V_{56}$

Table 6.111: $\text{Alt}_{13} < E_7, p = 2$

	V_{133}			V_{56}	
	1	32_a	64	12	32_b
1)	5	2	1	2	1

Modules tested: V_{133} , L_{132} , V_{912} , V_{56} .

Table 6.112: $\text{Alt}_{12} < E_7, p = 2$

		V_{133}					V_{56}			
		1	10	16_a	16_b	44	1	10	16_a	16_b
1)	N	5	2	2	2	1	4	2	1	1

Modules tested: V_{133} , L_{132} , V_{912} , V_{56} .

Table 6.113: $\text{Alt}_{11} < E_7, p = 2$

		V_{133}					V_{56}			
		1	10	16_a	16_b	44	1	10	16_a	16_b
1)		5	2	2	2	1	4	2	1	1

Modules tested: V_{133} , L_{132} , V_{912} , V_{56} .

Table 6.114: $\text{Alt}_{10} < E_7, p = 5$

		V_{133}				V_{56}
		28	35_a	35_b	35_c	28
1)	P	1	1	1	1	2

Modules tested: V_{133} , V_{912} , V_{56} .

Table 6.115: $\text{Alt}_{10} < E_7, p = 2$

		V_{133}				V_{56}			
		1	8	16	26	1	8	16	26
1)	N	5	1	1	4	4	0	0	2
2)	N	11	4	4	1	8	2	2	0

Modules tested: V133, L132, V912, V56.

Table 6.116: $\text{Alt}_9 < E_7, p = 0$ or $p \nmid |H|$

		V_{133}					V_{56}
		8	27	28	35_a	35_b	28
1)	P	1	1	1	1	1	2

Modules tested: V133, V912, V56.

Table 6.117: $\text{Alt}_9 < E_7, p = 7$

		V_{133}					V_{56}
		8	19	28	35_a	35_b	28
1)	P	2	1	1	1	1	2

Modules tested: V133, V912, V56.

Table 6.118: $\text{Alt}_9 < E_7, p = 5$

		V_{133}					V_{56}
		8	27	28	35_a	35_b	28
1)	P	1	1	1	1	1	2

Modules tested: V133, V912, V56.

Table 6.119: $\text{Alt}_9 < E_7, p = 3$

		V_{133}					V_{56}	
		1	7	21	27	35	7	21
1)	P, N	1	2	1	1	2	2	2

Modules tested: V133, V912, L856, V56.

Table 6.120: $\text{Alt}_9 < E_7, p = 2$

		V_{133}					V_{56}				
		1	8_a	8_b	8_c	26	1	8_a	8_b	8_c	26
1)	N	5	1	1	1	4	4	0	0	0	2
2)		11	4	4	4	1	8	2	2	2	0

Modules tested: V133, L132, V912, V56.

Table 6.121: $\text{Alt}_8 < E_7, p = 0$ or $p \nmid |H|$

		V_{133}					V_{56}	
		1	7	20	21_a	35	7	21_a
1)		1	3	1	1	2	2	2

Modules tested: V133, V912, V56.

Table 6.122: $\text{Alt}_8 < E_7, p = 7$

		V_{133}					V_{56}	
		1	7	19	21_a	35	7	21_a
1)		2	3	1	1	2	2	2

Modules tested: V133, V912, V56.

Table 6.123: $\text{Alt}_8 < E_7, p = 5$

		V_{133}								V_{56}		
		1	7	13	20	21_a	35	43	70	7	21_a	21_b
1)	P	0	0	0	1	0	0	1	1	2	0	2
2)		1	2	1	0	0	1	0	1	2	0	2
3)		1	3	0	1	1	2	0	0	2	2	0

Modules tested: V133, V912, V56.

Table 6.124: $\text{Alt}_8 < E_7$, $p = 3$

		V_{133}					V_{56}	
		1	7	13	21	35	7	21
1)	P, N	1	4	1	1	2	2	2

Modules tested: V133, V912, L856, V56.

Table 6.125: $\text{Alt}_7 < E_7$, $p = 0$ or $p \nmid |H|$

	V_{133}						V_{56}				
	1	6	10_a	10_b	14_a	15	1	6	10_a	10_b	15
1)	8	1	0	0	1	7	0	6	1	1	0
2)	4	5	2	2	1	3	2	4	0	0	2

Modules tested: V133, V912, V56.

Table 6.126: $\text{Alt}_7 < E_7$, $p = 7$

		V_{133}				V_{56}		
		1	5	10	14_a	1	5	10
1)	N	9	8	7	1	6	6	2

Modules tested: V133, V912, V56.

Table 6.127: $\text{Alt}_7 < E_7$, $p = 5$

		V_{133}								V_{56}						
		1	6	8	10_a	10_b	13	15	35	1	6	8	10_a	10_b	13	15
1)	P	0	3	7	1	1	3	0	0	0	0	2	2	2	0	0
2)		8	2	1	0	0	0	7	0	0	6	0	1	1	0	0
3)	N	1	2	3	0	0	2	0	2	2	2	2	0	0	2	0
4)		1	3	2	0	0	1	1	2	2	2	2	0	0	2	0
5)	N	4	1	2	1	1	4	0	1	2	2	2	0	0	2	0
6)	N	4	2	1	1	1	3	1	1	2	2	2	0	0	2	0
7)		4	6	1	2	2	0	3	0	2	4	0	0	0	0	2
8)		9	0	13	1	1	0	0	0	8	0	6	0	0	0	0

Modules tested: V133, V912, V56.

Table 6.128: $2 \cdot \text{Alt}_7 < E_7$, $p = 5$

		V_{133}				V_{56}			
		8	10_a	10_b	35	4_a	4_b	14_a	20_b
1)	P	1	1	1	3	1	1	2	1

Modules tested: V133, V912, V56.

Table 6.129: $\text{Alt}_7 < E_7$, $p = 3$

		V_{133}						V_{56}				
		1	6	10_a	10_b	13	15	1	6	10_a	10_b	15
1)		9	1	0	0	1	7	0	6	1	1	0
2)	N	5	5	2	2	1	3	2	4	0	0	2

Modules tested: V133, V912, L856, V56.

Table 6.130: $2 \cdot \text{Alt}_7 < E_7$, $p = 3$

		V_{133}				V_{56}			
		1	10_a	10_b	13	4_a	4_b	6_b	6_c
1)	N	22	1	1	7	1	1	7	1

Modules tested: V133, V912, L856, V56.

Table 6.131: $\text{Alt}_7 < E_7$, $p = 2$

		V_{133}						V_{56}					
		1	4_a	4_b	6	14	20	1	4_a	4_b	6	14	20
1)	N	15	0	0	1	8	0	0	1	1	8	0	0
2)	N	7	2	2	5	0	4	4	0	0	2	0	2
3)	N	7	2	2	6	1	3	4	0	0	2	0	2
4)	N	7	2	2	9	4	0	4	0	0	4	2	0
5)		19	8	8	5	0	1	12	4	4	2	0	0
6)		19	8	8	6	1	0	12	4	4	2	0	0

Modules tested: V133, L132, V912, V56.

Table 6.132: $\text{Alt}_6 < E_7$, $p = 0$ or $p \nmid |H|$

		V_{133}							V_{56}						
		1	5_a	5_b	8_a	8_b	9	10	1	5_a	5_b	8_a	8_b	9	10
1)	P	0	3	3	4	3	3	2	0	0	0	2	0	0	4
2)		1	5	2	3	2	3	3	0	0	0	2	0	0	4
3)		2	1	1	4	3	5	2	0	0	0	2	0	0	4
4)		2	4	4	2	1	3	4	0	0	0	2	0	0	4
5)		3	3	0	3	2	5	3	0	0	0	2	0	0	4
6)		4	2	2	2	1	5	4	0	0	0	2	0	0	4
7)		5	1	1	2	1	6	4	0	4	4	2	0	0	0
8)		3	4	1	4	2	3	3	1	2	2	0	1	3	0
9)		4	3	3	3	1	3	4	1	2	2	0	1	3	0
10)		1	2	2	4	3	4	2	2	2	2	2	0	2	0
11)		2	4	1	3	2	4	3	2	2	2	2	0	2	0
12)		3	3	3	2	1	4	4	2	2	2	2	0	2	0
13)		4	0	0	5	3	5	2	3	0	0	0	1	5	0
14)		6	1	1	3	1	5	4	3	0	0	0	1	5	0
15)		3	0	0	4	3	6	2	4	0	0	2	0	4	0
16)		5	1	1	2	1	6	4	4	0	0	2	0	4	0
17)		4	2	2	5	5	1	2	4	2	2	2	2	0	0
18)		5	4	1	4	4	1	3	4	2	2	2	2	0	0
19)		6	3	3	3	3	1	4	4	2	2	2	2	0	0
20)		6	0	0	5	5	3	2	6	0	0	2	2	2	0
21)		8	1	1	3	3	3	4	6	0	0	2	2	2	0
22)		9	9	0	0	0	1	7	6	6	0	0	0	0	2
23)		9	0	0	13	0	0	2	8	0	0	6	0	0	0

Memoir rows not reproduced: 7) (not compatible with V912); 9) (not compatible with V912).

Modules tested: V133, V912, V56.

Table 6.133: $2 \cdot \text{Alt}_6 < E_7$, $p = 0$ or $p \nmid |H|$

		V_{133}							V_{56}					
		1	5_a	5_b	8_a	8_b	9	10_a	4_a	4_b	8_c	8_d	10_b	10_c
1)	P	0	0	0	4	3	3	5	0	0	2	0	3	1
2)		2	1	1	2	1	3	7	0	0	2	0	3	1
3)		3	1	1	3	1	2	7	3	0	1	2	2	0
4)		1	0	0	5	3	2	5	1	1	0	1	3	1
5)		3	1	1	3	1	2	7	1	1	0	1	3	1

Modules tested: V133, V912, V56.

Table 6.134: $\text{Alt}_6 < E_7, p = 5$

		V_{133}					V_{56}				
		1	5_a	5_b	8	10	1	5_a	5_b	8	10
1)	P, N	3	3	3	10	2	0	0	0	2	4
2)	P, N	4	5	2	8	3	0	0	0	2	4
3)	P, N	5	4	4	6	4	0	0	0	2	4
4)	P, N	7	1	1	12	2	0	0	0	2	4
5)	P, N	8	3	0	10	3	0	0	0	2	4
6)	N	9	2	2	8	4	0	0	0	2	4
7)	P, N	9	0	0	13	2	0	4	4	2	0
8)	N	11	1	1	9	4	0	4	4	2	0
9)	N	5	2	2	11	2	4	2	2	4	0
10)	N	6	4	1	9	3	4	2	2	4	0
11)	N	7	3	3	7	4	4	2	2	4	0
12)		10	9	0	1	7	6	6	0	0	2
13)	N	9	0	0	13	2	8	0	0	6	0
14)	N	11	1	1	9	4	8	0	0	6	0

Modules tested: V133, V912, V56.

Table 6.135: $2 \cdot \text{Alt}_6 < E_7, p = 5$

		V_{133}					V_{56}			
		1	5_a	5_b	8	10_a	4_a	4_b	10_b	10_c
1)	P, N	3	0	0	10	5	2	2	3	1
2)	P, N	5	1	1	6	7	2	2	3	1
3)	P, N	5	1	1	6	7	6	3	2	0

Modules tested: V133, V912, V56.

Table 6.136: $\text{Alt}_5 < E_7, p = 0$ or $p \nmid |H|$

		V_{133}					V_{56}				
		1	3_a	3_b	4	5	1	3_a	3_b	4	5
1)		7	7	5	10	10	0	0	1	2	9
2)		8	10	6	3	13	0	9	1	4	2
3)		11	10	6	6	10	0	9	1	4	2
4)		3	6	5	8	13	0	6	4	4	2

	V_{133}					V_{56}				
	1	3_a	3_b	4	5	1	3_a	3_b	4	5
5)	6	6	5	11	10	0	6	4	4	2
6)	6	6	5	11	10	1	2	0	1	9
7)	6	7	7	5	13	2	6	6	2	2
8)	9	7	7	8	10	2	6	6	2	2
9)	4	7	5	7	13	3	0	1	5	6
10)	7	7	5	10	10	3	0	1	5	6
11)	3	6	5	8	13	4	2	0	4	6
12)	6	6	5	11	10	4	2	0	4	6
13)	17	22	0	0	10	4	14	0	0	2
14)	9	15	2	2	13	4	10	4	0	2
15)	12	15	2	5	10	4	10	4	0	2
16)	6	7	7	5	13	6	2	2	2	6
17)	9	7	7	8	10	6	2	2	2	6
18)	9	15	2	2	13	8	6	0	0	6
19)	12	15	2	5	10	8	6	0	0	6
20)	16	7	5	19	1	9	0	1	11	0
21)	15	6	5	20	1	10	2	0	10	0
22)	18	7	7	17	1	12	2	2	8	0
23)	21	15	2	14	1	14	6	0	6	0
24)	35	31	0	0	1	20	12	0	0	0

Modules tested: V_{133} , V_{912} , V_{56} .

Table 6.137: $2 \cdot \text{Alt}_5 < E_7$, $p = 0$ or $p \nmid |H|$

	V_{133}					V_{56}			
	1	3_a	3_b	4_a	5	2_a	2_b	4_b	6
1) P	0	9	8	8	10	0	2	4	6
2)	1	10	8	7	10	2	1	5	5
3)	3	15	5	5	10	5	0	7	3
4)	3	9	8	11	7	0	2	4	6
5)	3	9	8	11	7	3	5	1	6
6)	3	10	10	5	10	0	0	2	8
7)	4	10	8	10	7	2	1	5	5
8)	4	10	8	10	7	5	4	2	5
9)	6	15	5	8	7	5	0	7	3
10)	6	15	5	8	7	8	3	4	3
11)	6	10	10	8	7	0	0	2	8
12)	8	1	0	8	18	0	2	4	6
13)	8	13	9	6	7	0	8	1	6

	V_{133}					V_{56}			
	1	3_a	3_b	4_a	5	2_a	2_b	4_b	6
14)	9	2	0	7	18	2	1	5	5
15)	11	1	0	11	15	0	2	4	6
16)	11	1	0	11	15	3	5	1	6
17)	11	2	2	5	18	0	0	2	8
18)	12	2	0	10	15	2	1	5	5
19)	12	2	0	10	15	5	4	2	5
20)	13	18	9	1	7	7	0	9	1
21)	13	18	9	1	7	10	3	6	1
22)	14	28	0	0	7	14	0	7	0
23)	14	2	2	8	15	0	0	2	8
24)	16	5	1	6	15	0	8	1	6
25)	21	10	1	1	15	7	0	9	1
26)	21	10	1	1	15	10	3	6	1
27)	36	10	1	16	0	16	9	0	1
28)	52	27	0	0	0	26	0	1	0

Modules tested: V133, V912, V56.

Table 6.138: $\text{Alt}_5 < E_7$, $p = 3$

		V_{133}				V_{56}			
		1	3_a	3_b	4	1	3_a	3_b	4
1)	P, N	17	7	5	20	1	4	5	7
2)	N	21	10	6	16	2	9	1	6
3)	P, N	16	6	5	21	2	6	4	6
4)	N	19	7	7	18	4	6	6	4
5)	N	27	22	0	10	6	14	0	2
6)	N	22	15	2	15	6	10	4	2
7)	P, N	17	7	5	20	9	0	1	11
8)	N	16	6	5	21	10	2	0	10
9)	N	19	7	7	18	12	2	2	8
10)	N	22	15	2	15	14	6	0	6
11)	N	36	31	0	1	20	12	0	0

Modules tested: V133, V912, L856, V56.

Table 6.139: $2 \cdot \text{Alt}_5 < E_7$, $p = 3$

		V_{133}				V_{56}		
		1	3_a	3_b	4	2_a	2_b	6
1)	N	15	13	9	13	1	9	6
2)	N	31	5	1	21	1	9	6
3)		52	27	0	0	27	1	0
4)	P, N	13	10	10	15	2	2	8
5)	N	29	2	2	23	2	2	8
6)	P, N	10	9	8	18	4	6	6
7)	N	26	1	0	26	4	6	6
8)	P, N	11	10	8	17	7	6	5
9)	N	27	2	0	25	7	6	5
10)	P, N	13	15	5	15	12	7	3
11)	N	20	18	9	8	16	9	1
12)	N	36	10	1	16	16	9	1
13)	N	21	28	0	7	21	7	0

Modules tested: V133, V912, L856, V56.

Table 6.140: $M_{11} < E_7$, $p = 0$ or $p \nmid |H|$

		V_{133}							V_{56}				
		1	10_a	11	16_a	16_b	45	55	1	10_a	11	16_a	16_b
1)		2	0	2	2	2	1	0	2	0	2	1	1
2)		3	0	1	2	2	0	1	2	0	2	1	1
3)		4	2	0	2	2	1	0	4	2	0	1	1

Modules tested: V133, V912, V56.

Table 6.141: $M_{11} < E_7$, $p = 11$

		V_{133}							V_{56}			
		1	9	10_a	10_b	11	16	55	1	9	11	16
1)		2	1	1	1	2	5	0	2	0	2	2
2)		3	0	0	0	1	4	1	2	0	2	2
3)	N	6	3	1	1	0	5	0	6	2	0	2

Modules tested: V133, V912, V56.

Table 6.142: $M_{11} < E_7$, $p = 5$

		V_{133}							V_{56}				
		1	10_a	11	16_a	16_b	45	55	1	10_a	11	16_a	16_b
1)		1	1	2	0	0	1	1	2	0	2	1	1
2)	$\mathbf{N}$	2	0	2	2	2	1	0	2	0	2	1	1
3)		2	1	1	0	0	0	2	2	0	2	1	1
4)	$\mathbf{N}$	3	0	1	2	2	0	1	2	0	2	1	1
5)		3	3	0	0	0	1	1	4	2	0	1	1
6)	$\mathbf{N}$	4	2	0	2	2	1	0	4	2	0	1	1

Modules tested: V133, V912, V56.

Table 6.143: $M_{11} < E_7$, $p = 3$

		V_{133}								V_{56}					
		1	5_a	5_b	10_a	10_b	10_c	24	45	1	5_a	5_b	10_a	10_b	10_c
1)	N	8	2	2	2	2	2	0	1	6	1	1	2	1	1
2)	N	8	4	4	0	2	2	0	1	6	3	3	0	1	1
3)	N	9	4	4	0	3	3	1	0	6	3	3	0	1	1

Modules tested: V133, V912, L856, V56.

Table 6.144: $M_{11} < E_7$, $p = 2$

		V_{133}					V_{56}			
		1	10	16_a	16_b	44	1	10	16_a	16_b
1)	N	5	2	2	2	1	4	2	1	1
2)	N	7	5	1	1	1	4	2	1	1

Modules tested: V133, L132, V912, V56.

Table 6.145: $2 \cdot M_{12} < E_7$, $p = 0$ or $p \nmid |H|$

		V_{133}				V_{56}	
		1	16_a	16_b	66	12	32
1)		3	2	2	1	2	1

Modules tested: V133, V912, V56.

Table 6.146: $2 \cdot M_{12} < E_7$, $p = 11$

	V_{133}			V_{56}	
	1	16	66	12	32
1)	3	4	1	2	1

Modules tested: V133, V912, V56.

Table 6.147: $M_{12} < E_7$, $p = 5$

	V_{133}					V_{56}			
	1	11_a	16_a	16_b	78	1	11_a	16_a	16_b
1)	1	2	1	1	1	2	2	1	1

Modules tested: V133, V912, V56.

Table 6.148: $2 \cdot M_{12} < E_7$, $p = 5$

		V_{133}						V_{56}	
		1	16_a	16_b	55_a	66	78	12	32
1)	P	0	0	0	1	0	1	2	1
2)		3	2	2	0	1	0	2	1

Modules tested: V133, V912, V56.

Table 6.149: $2 \cdot M_{12} < E_7$, $p = 3$

		V_{133}				V_{56}			
		1	15_a	15_b	34	6_a	6_b	10_c	10_d
1)	N	9	3	3	1	3	3	1	1

Modules tested: V133, V912, L856, V56.

Table 6.150: $M_{12} < E_7$, $p = 2$

		V_{133}					V_{56}			
		1	10	16_a	16_b	44	1	10	16_a	16_b
1)	N	5	2	2	2	1	4	2	1	1

Modules tested: V133, L132, V912, V56.

Table 6.151: $2 \cdot M_{22} < E_7$, $p = 5$

		V_{133}	V_{56}	
		133	28_a	28_b
1)	P	1	1	1

Modules tested: V133, V912, V56.

Table 6.152: $J_1 < E_7$, $p = 11$

		V_{133}					V_{56}			
		1	7	14	27	64	1	7	14	27
1)		3	5	1	3	0	0	4	2	0
2)		1	0	1	2	1	2	0	0	2
3)		21	14	1	0	0	14	6	0	0

Modules tested: V133, V912, V56.

Table 6.153: $2 \cdot J_2 < E_7$, $p = 0$ or $p \nmid |H|$

		V_{133}			V_{56}	
		1	14_a	21_b	6_a	14_c
1)		14	7	1	7	1

Modules tested: V133, V912, V56.

Table 6.154: $2 \cdot J_2 < E_7$, $p = 7$

		V_{133}			V_{56}	
		1	14_a	21_b	6_a	14_c
1)		14	7	1	7	1

Modules tested: V133, V912, V56.

Table 6.155: $2 \cdot J_2 < E_7, p = 5$

		V_{133}			V_{56}	
		1	14_a	21	6	14_b
1)		14	7	1	7	1

Modules tested: V133, V912, V56.

Table 6.156: $2 \cdot J_2 < E_7, p = 3$

		V_{133}			V_{56}	
		1	13_a	21_a	6_b	14
1)	N	21	7	1	7	1

Modules tested: V133, V912, L856, V56.

Table 6.157: $J_2 < E_7, p = 2$

		V_{133}							V_{56}				
		1	6_a	6_b	14_a	14_b	64_b	84	1	6_a	6_b	14_a	36
1)	P	1	1	0	1	2	0	1	2	1	2	0	1
2)		15	1	0	0	8	0	0	2	1	8	0	0
3)	N	11	8	3	4	0	0	0	4	4	0	2	0
4)	N	3	2	2	3	0	1	0	4	2	2	2	0
5)	N	35	14	0	1	0	0	0	20	6	0	0	0

Modules tested: V133, L132, V912, V56.

Table 6.158: $2 \cdot Ru < E_7, p = 5$

		V_{133}	V_{56}	
		133	28_a	28_b
1)	P	1	1	1

Modules not used (elements of large order): V912.

Modules tested: V133, V56.

Table 6.159: $2 \cdot HS < E_7$, $p = 5$

		V_{133}	V_{56}	
		133_a	28_a	28_b
1)	P	1	1	1

Modules tested: V133, V912, V56.

Table 6.160: $L_2(7) < E_7$, $p = 0$ or $p \nmid |H|$

		V_{133}						V_{56}					
		1	3_a	3_b	6	7	8	1	3_a	3_b	6	7	8
1)	P	0	1	1	6	5	7	0	2	2	0	4	2
2)		1	2	2	6	4	7	2	0	0	4	2	2
3)		2	3	3	6	3	7	0	2	2	0	4	2
4)		3	1	1	6	8	4	0	2	2	0	4	2
5)		3	4	4	6	2	7	4	2	2	4	0	2
6)		4	2	2	6	7	4	2	0	0	4	2	2
7)		5	3	3	6	6	4	0	2	2	0	4	2
8)		6	4	4	6	5	4	4	2	2	4	0	2
9)		8	0	0	2	7	8	0	1	1	6	2	0
10)		9	1	1	0	2	13	8	0	0	0	0	6
11)		10	2	2	2	5	8	2	3	3	6	0	0
12)		11	9	9	6	0	4	4	6	6	0	0	2
13)		12	1	1	0	5	10	8	0	0	0	0	6
14)		15	7	7	2	0	8	2	7	7	2	0	0
15)		21	1	1	0	14	1	14	0	0	0	6	0
16)		35	15	15	0	0	1	20	6	6	0	0	0

Memoir rows not reproduced: 7) (not compatible with V912); 10) (not compatible with V912).

Modules tested: V133, V912, V56.

Table 6.161: $SL_2(7) < E_7$, $p = 0$ or $p \nmid |H|$

		V_{133}						V_{56}				
		1	3_a	3_b	6_a	7	8_a	4_a	4_b	6_b	6_c	8_b
1)	P	0	4	4	3	5	7	2	2	3	1	2
2)		3	4	4	3	8	4	2	2	3	1	2
3)		14	1	1	7	1	8	1	1	7	1	0

Modules tested: V133, V912, V56.

Table 6.162: $L_2(7) < E_7, p = 3$

		V_{133}					V_{56}				
		1	3 _a	3 _b	6	7	1	3 _a	3 _b	6	7
1)	P, N	7	1	1	6	12	2	2	2	0	6
2)	N	8	2	2	6	11	4	0	0	4	4
3)	P, N	9	3	3	6	10	2	2	2	0	6
4)	N	10	4	4	6	9	6	2	2	4	2
5)	N	15	9	9	6	4	6	6	6	0	2
6)	N	16	0	0	2	15	0	1	1	6	2
7)	N	18	2	2	2	13	2	3	3	6	0
8)	N	22	1	1	0	15	14	0	0	0	6
9)	N	23	7	7	2	8	2	7	7	2	0
10)	N	36	15	15	0	1	20	6	6	0	0

Modules tested: V133, V912, L856, V56.

Table 6.163: $SL_2(7) < E_7, p = 3$

		V_{133}					V_{56}			
		1	3 _a	3 _b	6 _a	7	4 _a	4 _b	6 _b	6 _c
1)	P, N	7	4	4	3	12	4	4	3	1
2)	P, N	22	1	1	7	9	1	1	7	1

Modules tested: V133, V912, L856, V56.

Table 6.164: $L_2(8) < E_7, p = 0$ or $p \nmid |H|$

		V_{133}									V_{56}								
		1	7 _a	7 _b	7 _c	7 _d	8	9 _a	9 _b	9 _c	1	7 _a	7 _b	7 _c	7 _d	8	9 _a	9 _b	9 _c
1)		1	2	1	1	1	2	3	3	3	0	2	2	2	2	0	0	0	0
2)		1	2	1	1	1	2	3	3	3	2	0	0	0	0	0	2	2	2
3)		3	1	3	2	1	0	3	3	3	0	2	3	1	2	0	0	0	0
4)		3	1	5	1	0	0	3	3	3	0	2	4	2	0	0	0	0	0
5)		3	2	1	1	1	4	4	0	3	4	0	0	0	0	2	2	0	2
6)		5	2	1	1	1	6	4	0	1	3	0	0	0	0	1	5	0	0
7)		6	5	1	1	1	1	4	0	3	6	2	0	0	0	0	2	0	2
8)		10	2	1	1	1	11	0	0	0	8	0	0	0	0	6	0	0	0
9)		13	5	1	1	1	8	0	0	0	10	2	0	0	0	4	0	0	0
10)		21	1	14	1	0	0	0	0	0	14	0	6	0	0	0	0	0	0

Memoir rows not reproduced: 3) (not compatible with V912).
Modules tested: V133, V912, V56.

Table 6.165: $L_2(8) < E_7, p = 7$

		V_{133}						V_{56}					
		1	7_a	7_b	7_c	7_d	8	1	7_a	7_b	7_c	7_d	8
1)	P, N	10	2	1	1	1	11	0	2	2	2	2	0
2)	N	10	2	1	1	1	11	8	0	0	0	0	6
3)	N	12	1	5	1	0	9	0	2	4	2	0	0
4)	N	12	1	3	2	1	9	0	2	3	1	2	0
5)	N	13	2	5	1	0	8	0	5	3	0	0	0
6)	N	13	5	1	1	1	8	10	2	0	0	0	4
7)		21	1	14	1	0	0	14	0	6	0	0	0

Modules tested: V133, V912, V56.

Table 6.166: $L_2(8) < E_7, p = 3$

		V_{133}					V_{56}				
		1	7	9_a	9_b	9_c	1	7	9_a	9_b	9_c
1)	P, N	3	7	3	3	3	0	8	0	0	0
2)	N	3	7	3	3	3	2	0	2	2	2
3)	N	5	8	3	3	2	2	0	1	3	2
4)	N	7	9	4	0	3	6	2	2	0	2
5)	N	11	11	4	0	1	4	1	5	0	0
6)	N	21	16	0	0	0	14	6	0	0	0

Modules tested: V133, V912, L856, V56.

Table 6.167: $L_2(11) < E_7, p = 0$ or $p \nmid |H|$

	V_{133}								V_{56}						
	1	5_a	5_b	10_a	10_b	11	12_a	12_b	1	5_a	5_b	10_a	10_b	11	12_a
1)	1	2	2	1	2	2	3	2	2	1	1	0	2	0	2
2)	2	0	0	2	4	1	3	2	2	1	1	0	2	0	2
3)	4	2	2	1	2	5	1	1	4	1	1	0	2	2	0
4)	5	0	0	2	4	4	1	1	4	1	1	0	2	2	0
5)	9	4	4	6	0	0	1	1	6	3	3	2	0	0	0

Modules tested: V133, V912, V56.

Table 6.168: $SL_2(11) < E_7$, $p = 0$ or $p \nmid |H|$

		V_{133}								V_{56}					
		1	5_a	5_b	10_a	10_b	11	12_a	12_b	6_a	6_b	10_d	10_e	12_c	12_d
1)	P	0	1	1	3	0	3	3	2	1	1	1	1	2	0
2)		1	1	1	3	0	4	3	1	0	0	1	1	1	2
3)		3	1	1	3	0	6	1	1	3	3	1	1	0	0
4)		8	3	3	0	6	1	1	1	3	3	1	1	0	0

Modules tested: V133, V912, V56.

Table 6.169: $L_2(11) < E_7$, $p = 5$

		V_{133}						V_{56}					
		1	5_a	5_b	10_a	10_b	11	1	5_a	5_b	10_a	10_b	11
1)	N	6	2	2	1	2	7	4	1	1	0	2	2
2)	N	7	0	0	2	4	6	4	1	1	0	2	2
3)	N	11	4	4	6	0	2	6	3	3	2	0	0

Modules tested: V133, V912, V56.

Table 6.170: $SL_2(11) < E_7$, $p = 5$

		V_{133}						V_{56}			
		1	5_a	5_b	10_a	10_b	11	6_a	6_b	10_d	10_e
1)	P, N	5	1	1	3	0	8	3	3	1	1
2)	N	10	3	3	0	6	3	3	3	1	1

Modules tested: V133, V912, V56.

Table 6.171: $L_2(11) < E_7$, $p = 3$

		V_{133}						V_{56}				
		1	5_a	5_b	10	12_a	12_b	1	5_a	5_b	10	12_a
1)	N	3	4	4	3	3	2	2	3	3	0	2
2)	N	9	4	4	6	1	1	6	3	3	2	0

Modules tested: V133, V912, L856, V56.

Table 6.172: $SL_2(11) < E_7, p = 3$

		V_{133}						V_{56}				
		1	5_a	5_b	10_a	12_a	12_b	6_a	6_b	10_b	12_c	12_d
1)	P, N	3	1	1	6	3	2	1	1	2	2	0
2)	P, N	5	1	1	7	3	1	0	0	2	1	2
3)	N	9	1	1	9	1	1	3	3	2	0	0
4)		9	9	9	1	1	1	3	3	2	0	0

Modules tested: V133, V912, L856, V56.

Table 6.173: $L_2(11) < E_7, p = 2$

		V_{133}						V_{56}					
		1	5_a	5_b	10	12_a	12_b	1	5_a	5_b	10	12_a	12_b
1)	N	3	1	1	6	3	2	2	1	1	2	2	0
2)	N	3	4	4	3	3	2	2	1	1	2	2	0
3)	N	5	2	2	6	3	1	0	0	0	2	1	2
4)	P, N	5	5	5	3	3	1	0	0	0	2	1	2
5)	N	9	4	4	6	1	1	6	3	3	2	0	0
6)	N	9	7	7	3	1	1	6	3	3	2	0	0

Modules tested: V133, L132, V912, V56.

Table 6.174: $L_2(13) < E_7, p = 0$ or $p \nmid |H|$

		V_{133}									V_{56}						
		1	7_a	7_b	12_a	12_b	12_c	13	14_a	14_b	1	7_a	7_b	12_a	13	14_a	14_b
1)		1	0	0	1	1	1	2	3	2	0	2	2	0	0	0	2
2)		1	0	0	1	1	1	2	3	2	2	0	0	0	2	2	0
3)		2	2	2	1	1	1	3	2	0	0	2	2	0	0	0	2
4)		2	2	2	1	1	1	3	2	0	2	0	0	0	2	2	0
5)		3	0	0	4	1	0	0	3	2	4	0	0	2	0	2	0
6)		3	5	0	0	0	0	3	3	1	0	4	0	0	0	0	2
7)		4	2	2	4	1	0	1	2	0	4	0	0	2	0	2	0
8)		21	14	0	0	0	0	0	0	1	14	6	0	0	0	0	0

Modules tested: V133, V912, V56.

Table 6.175: $SL_2(13) < E_7$, $p = 0$ or $p \nmid |H|$

		V_{133}									V_{56}						
		1	7_a	7_b	12_a	12_b	12_c	13	14_a	14_b	6_a	12_d	12_e	12_f	14_c	14_d	14_e
1)	P	0	2	0	1	1	1	1	1	4	1	1	1	1	1	0	0
2)	P	0	1	1	1	1	1	1	1	4	0	0	0	0	2	1	1
3)		1	2	0	2	1	1	0	1	4	1	1	2	0	1	0	0
4)		1	4	2	1	1	1	2	0	2	1	1	1	1	1	0	0
5)		1	3	3	1	1	1	2	0	2	0	0	0	0	2	1	1
6)		2	4	2	2	1	1	1	0	2	1	1	2	0	1	0	0
7)		14	1	0	0	0	0	0	7	1	7	0	0	0	1	0	0

Modules tested: V133, V912, V56.

Table 6.176: $L_2(13) < E_7$, $p = 7$

		V_{133}						V_{56}					
		1	7_a	7_b	12	14_a	14_b	1	7_a	7_b	12	14_a	14_b
1)	P, N	3	0	0	5	3	2	0	2	2	0	0	2
2)	N	3	0	0	5	3	2	4	0	0	2	2	0
3)	P, N	5	2	2	6	2	0	0	2	2	0	0	2
4)	N	5	2	2	6	2	0	4	0	0	2	2	0
5)	N	6	5	0	3	3	1	0	4	0	0	0	2
6)		21	14	0	0	0	1	14	6	0	0	0	0

Modules tested: V133, V912, V56.

Table 6.177: $SL_2(13) < E_7$, $p = 7$

		V_{133}						V_{56}				
		1	7_a	7_b	12	14_a	14_b	6_a	6_b	14_c	14_d	14_e
1)	P, N	1	2	0	4	1	4	4	3	1	0	0
2)	P, N	1	1	1	4	1	4	0	0	2	1	1
3)	P, N	3	4	2	5	0	2	4	3	1	0	0
4)	P, N	3	3	3	5	0	2	0	0	2	1	1
5)		14	1	0	0	7	1	7	0	1	0	0

Modules tested: V133, V912, V56.

Table 6.178: $L_2(13) < E_7, p = 3$

		V_{133}							V_{56}				
		1	7_a	7_b	12_a	12_b	12_c	13	1	7_a	7_b	12_a	13
1)	P, N	4	2	2	1	1	1	5	0	4	4	0	0
2)	N	4	2	2	1	1	1	5	4	0	0	0	4
3)	N	6	6	1	0	0	0	6	0	6	2	0	0
4)	N	6	2	2	4	1	0	3	6	0	0	2	2
5)		21	15	1	0	0	0	0	14	6	0	0	0

Modules tested: V133, V912, L856, V56.

Table 6.179: $SL_2(13) < E_7, p = 3$

		V_{133}							V_{56}				
		1	7_a	7_b	12_a	12_b	12_c	13	6_a	12_d	12_e	12_f	14
1)	P, N	1	6	4	1	1	1	2	1	1	1	1	1
2)	P, N	1	5	5	1	1	1	2	0	0	0	0	4
3)		2	6	4	2	1	1	1	1	1	2	0	1
4)	N	21	2	1	0	0	0	7	7	0	0	0	1

Modules tested: V133, V912, L856, V56.

Table 6.180: $L_2(13) < E_7, p = 2$

		V_{133}							V_{56}							
		1	6_a	6_b	12_a	12_b	12_c	14	1	6_a	6_b	12_a	12_b	12_c	14	
1)		3	0	0	4	1	0	5	4	0	0	2	0	0	2	
2)		N	3	2	0	2	1	1	5	2	1	2	1	2	0	0
3)		P, N	3	3	1	1	1	1	5	2	1	2	1	1	1	0
4)		N	3	2	2	1	1	1	5	4	2	2	0	0	0	2
5)		N	5	5	0	0	0	0	7	4	4	0	0	0	0	2
6)		N	9	3	3	4	1	0	2	4	0	0	2	0	0	2
7)		N	9	5	3	2	1	1	2	2	1	2	1	2	0	0
8)		P, N	9	6	4	1	1	1	2	2	1	2	1	1	1	0
9)		N	9	5	5	1	1	1	2	4	2	2	0	0	0	2
10)		N	11	8	3	0	0	0	4	4	4	0	0	0	0	2
11)		N	15	1	0	0	0	0	8	2	1	8	0	0	0	0
12)		N	35	14	0	0	0	0	1	20	6	0	0	0	0	0

Modules tested: V133, L132, V912, V56.

Table 6.181: $L_2(17) < E_7$, $p = 0$ or $p \nmid |H|$

	V_{133}							V_{56}					
	1	9 _a	16 _a	17	18 _a	18 _b	18 _c	1	9 _a	9 _b	16 _a	17	18 _a
1)	1	3	1	1	2	1	1	2	2	0	0	0	2
2)	3	3	1	3	0	1	1	2	4	2	0	0	0
3)	3	3	1	3	0	1	1	4	2	0	0	2	0
4)	6	3	4	0	0	1	1	6	2	0	2	0	0

Modules tested: V133, V912, V56.

Table 6.182: $L_2(17) < E_7$, $p = 3$

		V_{133}						V_{56}				
		1	9_a	16	18_a	18_b	18_c	1	9_a	9_b	16	18_a
1)	N	2	3	2	2	1	1	2	2	0	0	2
2)	N	6	3	4	0	1	1	2	4	2	0	0
3)	N	6	3	4	0	1	1	6	2	0	2	0

Modules tested: V133, V912, L856, V56.

Table 6.183: $L_2(17) < E_7$, $p = 2$

		V_{133}				V_{56}			
		1	8 _a	8 _b	16 _a	1	8 _a	8 _b	16 _a
1)	N	13	5	2	4	8	2	0	2
2)	N	13	8	5	1	8	4	2	0

Modules tested: V133, L132, V912, V56.

Table 6.184: $L_2(19) < E_7$, $p = 0$ or $p \nmid |H|$

	V_{133}								V_{56}			
	1	9 _a	9 _b	18 _a	18 _d	20 _b	20 _c	20 _d	1	9 _a	9 _b	18 _d
1)	1	1	1	1	2	1	1	1	2	1	1	2

Modules tested: V133, V912, V56.

Table 6.185: $SL_2(19) < E_7$, $p = 0$ or $p \nmid |H|$

		V_{133}						V_{56}				
		18_a	18_b	19	20_b	20_c	20_d	10_a	10_b	18_f	18_h	20_e
1)	P	2	1	1	1	1	1	1	1	1	1	0
2)	P	2	1	1	1	1	1	0	0	1	1	1

Modules not used (elements of large order): V912.

Modules tested: V133, V56.

Table 6.186: $L_2(19) < E_7$, $p = 5$

	V_{133}							V_{56}		
	1	9_a	9_b	18	20_b	20_c	20_d	1	9_a	9_b
1)	1	3	3	1	1	1	1	2	3	3

Modules tested: V133, V912, V56.

Table 6.187: $SL_2(19) < E_7$, $p = 5$

		V_{133}					V_{56}			
		1	18_a	20_b	20_c	20_d	10_a	10_b	18_b	20_e
1)	P, N	1	4	1	1	1	0	0	2	1
2)	P, N	1	4	1	1	1	1	1	2	0

Modules not used (elements of large order): V912.

Modules tested: V133, V56.

Table 6.188: $L_2(19) < E_7$, $p = 3$

		V_{133}						V_{56}			
		1	9_a	9_b	18_a	18_d	19	1	9_a	9_b	18_d
1)	N	4	1	1	1	2	3	2	1	1	2

Modules tested: V133, V912, L856, V56.

Table 6.189: $SL_2(19) < E_7, p = 3$

		V_{133}				V_{56}			
		1	18_a	18_b	19	10_a	10_b	18_f	18_h
1)	P, N	3	2	1	4	1	1	1	1

Modules not used (elements of large order): V912, L856.

Modules tested: V133, V56.

Table 6.190: $L_2(19) < E_7, p = 2$

		1	9_a	9_b	18_a	18_b	20_b	20_c	20_d	1	9_a	9_b	18_a	18_b	20_a
		1	0	0	3	1	1	1	1	0	1	1	0	1	1
1)		1	0	0	3	1	1	1	1	2	2	2	0	1	0
2)		1	0	0	3	1	1	1	1	0	0	0	2	0	1
3)	P, N	1	1	1	2	1	1	1	1	2	1	1	2	0	0
4)	N	1	1	1	2	1	1	1	1	2	1	1	2	0	0

Modules tested: V133, L132, V912, V56.

Table 6.191: $L_2(25) < E_7, p = 0$ or $p \nmid |H|$

		1	25	26_a	26_b	26_c	26_d	26_e	1	26_a	26_c
		3	0	3	0	0	1	1	4	2	0
1)		3	0	3	0	0	1	1	4	2	0
2)		4	1	2	2	0	0	0	4	2	0
3)		4	1	0	2	2	0	0	4	0	2

Modules tested: V133, V912, V56.

Table 6.192: $L_2(25) < E_7, p = 13$

		1	24	26_a	26_b	26_c	26_d	26_e	1	26_a	26_c
		3	0	3	0	0	1	1	4	2	0
1)		3	0	3	0	0	1	1	4	2	0
2)		5	1	2	2	0	0	0	4	2	0
3)		5	1	0	2	2	0	0	4	0	2

Modules tested: V133, V912, V56.

Table 6.193: $L_2(25) < E_7, p = 3$

		V_{133}					V_{56}			
		1	13_a	13_b	25	26	1	13_a	13_b	25
1)		4	2	2	1	2	4	2	2	0
2)	N	6	0	0	3	2	6	0	0	2

Modules tested: V133, V912, L856, V56.

Table 6.194: $L_2(25) < E_7, p = 2$

		V_{133}				V_{56}	
		1	12_a	12_b	26	1	26
1)		3	0	0	5	4	2
2)	N	9	3	3	2	4	2

Modules tested: V133, L132, V912, V56.

Table 6.195: $L_2(27) < E_7, p = 0$ or $p \nmid |H|$

		V_{133}				V_{56}	
		1	26_a	26_b	26_e	1	26_e
1)		3	1	1	3	4	2

Modules tested: V133, V912, V56.

Table 6.196: $SL_2(27) < E_7, p = 0$ or $p \nmid |H|$

		V_{133}						V_{56}			
		1	26_a	26_b	26_c	27	28_a	14_a	14_b	28_j	28_k
1)	P	0	1	1	1	1	1	0	0	1	1
2)		1	1	1	1	2	0	2	2	0	0

Modules tested: V133, V912, V56.

Table 6.197: $L_2(27) < E_7$, $p = 13$

		V_{133}				V_{56}	
		1	26_a	26_b	26_e	1	26_e
1)		3	1	1	3	4	2

Modules tested: V133, V912, V56.

Table 6.198: $SL_2(27) < E_7$, $p = 13$

		V_{133}					V_{56}	
		1	26_a	26_b	26_c	27	14_a	14_b
1)	P, N	1	1	1	1	2	2	2

Modules tested: V133, V912, V56.

Table 6.199: $L_2(27) < E_7$, $p = 7$

		V_{133}				V_{56}		
		1	13_a	13_b	26	1	13_a	13_b
1)	N	3	3	3	2	4	2	2

Modules tested: V133, V912, V56.

Table 6.200: $SL_2(27) < E_7$, $p = 7$

		V_{133}				V_{56}			
		1	26_a	28_a		14_a	14_b	28_j	28_k
1)	P, N	1	4	1		0	0	1	1
2)	P, N	3	5	0		2	2	0	0

Modules tested: V133, V912, V56.

Table 6.201: $L_2(27) < E_7$, $p = 2$

		V_{133}							V_{56}					
		1	13_a	13_b	26_a	26_b	26_c	28_a	1	13_a	13_b	26_a	28_d	28_e
1)	P, N	1	1	1	1	1	1	1	0	0	0	0	1	1
2)		3	0	0	4	0	1	0	4	0	0	2	0	0

		V_{133}							V_{56}					
		1	13_a	13_b	26_a	26_b	26_c	28_a	1	13_a	13_b	26_a	28_d	28_e
3)	N	3	2	2	1	1	1	0	4	2	2	0	0	0

Modules tested: V133, L132, V912, V56.

Table 6.202: $SL_2(29) < E_7$, $p = 0$ or $p \nmid |H|$

		V_{133}					V_{56}	
		15_a	28_b	30_d	30_e	30_f	28_k	28_m
1)	P	1	1	1	1	1	1	1

Modules not used (elements of large order): V912.

Modules tested: V133, V56.

Table 6.203: $SL_2(29) < E_7$, $p = 7$

		V_{133}			V_{56}	
		15_a	15_b	28_b	28_k	28_m
1)	P	4	3	1	1	1

Modules not used (elements of large order): V912.

Modules tested: V133, V56.

Table 6.204: $SL_2(29) < E_7$, $p = 5$

		V_{133}					V_{56}
		15_a	28_b	30_d	30_e	30_f	28_c
1)	P	1	1	1	1	1	2

Modules not used (elements of large order): V912.

Modules tested: V133, V56.

Table 6.205: $SL_2(29) < E_7$, $p = 3$

		V_{133}					V_{56}
		15_a	28_b	30_d	30_e	30_f	28_d
1)	P	1	1	1	1	1	2

Modules not used (elements of large order): V912, L856.
Modules tested: V133, V56.

Table 6.206: $L_2(29) < E_7, p = 2$

		V_{133}						V_{56}	
		1	14_a	28_b	30_a	30_b	30_c	28_d	28_f
1)	P	1	1	1	1	1	1	1	1

Modules tested: V133, L132, V912, V56.

Table 6.207: $SL_2(37) < E_7, p = 0$ or $p \nmid |H|$

Not recomputed: no character table available in GAP. See the Memoir.

Table 6.208: $SL_2(37) < E_7, p = 19$

Not recomputed: no character table available in GAP. See the Memoir.

Table 6.209: $SL_2(37) < E_7, p = 3$

Not recomputed: no character table available in GAP. See the Memoir.

Table 6.210: $L_2(37) < E_7, p = 2$

Not recomputed: no Brauer table available in GAP. See the Memoir.

Table 6.211: $L_3(3) < E_7, p = 0$ or $p \nmid |H|$

	V_{133}												V_{56}					
	1	13	16_a	16_b	16_c	16_d	26_a	26_b	26_c	27	39	1	12	16_c	16_d	26_a	27	
1)	1	0	0	0	0	0	1	1	1	2	0	2	0	0	0	0	2	
2)	2	1	2	2	0	0	0	0	0	2	0	0	2	1	1	0	0	
3)	2	1	1	1	1	1	0	0	0	2	0	2	0	0	0	0	2	
4)	3	0	0	0	0	0	3	1	1	0	0	4	0	0	0	2	0	
5)	3	0	2	2	0	0	0	0	0	1	1	0	2	1	1	0	0	
6)	3	0	1	1	1	1	0	0	0	1	1	2	0	0	0	0	2	

	V ₁₃₃											V ₅₆					
	1	13	16 _a	16 _b	16 _c	16 _d	26 _a	26 _b	26 _c	27	39	1	12	16 _c	16 _d	26 _a	27
7)	4	1	1	1	1	1	2	0	0	0	0	4	0	0	0	2	0

Modules tested: V133, V912, V56.

Table 6.212: $L_3(3) < E_7, p = 13$

		V_{133}							V_{56}				
		1	11	13	16	26_a	26_b	26_c	39	1	11	16	26_a
1)	N	1	2	0	2	1	1	1	0	2	2	2	0
2)	N	2	2	1	6	0	0	0	0	2	2	2	0
3)		3	0	0	0	3	1	1	0	4	0	0	2
4)		3	1	0	5	0	0	0	1	2	2	2	0
5)		4	0	1	4	2	0	0	0	4	0	0	2

Modules tested: V133, V912, V56.

Table 6.213: $L_3(3) < E_7, p = 2$

		V_{133}							V_{56}				
		1	12	16_a	16_b	16_c	16_d	26	1	12	16_c	16_d	26
1)	N	3	0	0	0	0	0	5	4	0	0	0	2
2)	N	5	1	2	2	0	0	2	0	2	1	1	0
3)	N	5	1	1	1	1	1	2	4	0	0	0	2

Modules tested: V133, L132, V912, V56.

Table 6.214: $2 \cdot L_3(4) < E_7, p = 0$ or $p \nmid |H|$

		V ₁₃₃			V ₅₆
		35 _b	35 _c	63 _a	28 _a
1)	P	1	1	1	2

Modules tested: V133, V912, V56.

Table 6.215: $2 \cdot L_3(4) < E_7, p = 7$

		V_{133}			V_{56}
		35_b	35_c	63_a	28_a
1)	P	1	1	1	2

Modules tested: V133, V912, V56.

Table 6.216: $2 \cdot L_3(4) < E_7, p = 5$

		V_{133}			V_{56}
		35_b	35_c	63	28
1)	P	1	1	1	2

Modules tested: V133, V912, V56.

Table 6.217: $2 \cdot L_3(4) < E_7, p = 3$

		V_{133}						V_{56}			
		1	15_a	15_b	15_c	19	63_a	6	10_a	10_b	22_b
1)	P, N	2	0	1	1	2	1	2	0	0	2
2)		9	7	0	0	1	0	6	1	1	0

Modules tested: V133, V912, L856, V56.

Table 6.218: $L_4(3) < E_7, p = 2$

		V_{133}			V_{56}	
		1	26_a	26_b	1	26_a
1)		3	4	1	4	2

Modules tested: V133, L132, V912, V56.

Table 6.219: $U_3(3) < E_7, p = 0$ or $p \nmid |H|$

		V_{133}														V_{56}											
		1	6	7_a	7_b	7_c	14	21_a	21_b	21_c	27	28_a	28_b	32_a	32_b	1	6	7_a	7_b	7_c	14	21_a	21_b	21_c	27	28_a	28_b
1)	P	0	0	0	0	0	0	0	1	1	1	0	0	1	1	0	0	0	0	0	0	0	0	0	0	1	1
2)		21	0	14	0	0	1	0	0	0	0	0	0	0	0	14	0	6	0	0	0	0	0	0	0	0	0

		V ₁₃₃													V ₅₆												
		1	6	7 _a	7 _b	7 _c	14	21 _a	21 _b	21 _c	27	28 _a	28 _b	32 _a	32 _b	1	6	7 _a	7 _b	7 _c	14	21 _a	21 _b	21 _c	27	28 _a	28 _b
3)	new	14	0	0	0	0	7	1	0	0	0	0	0	0	0	0	7	0	1	1	0	0	0	0	0	0	0
4)	new	6	6	0	2	2	3	1	0	0	0	0	0	0	0	0	4	4	0	0	0	2	0	0	0	0	0
5)	new	3	0	5	0	0	1	0	0	0	3	0	0	0	0	0	0	0	4	0	0	2	0	0	0	0	0
6)	new	2	0	1	1	1	0	0	0	0	2	1	1	0	0	0	2	0	0	0	0	0	0	0	2	0	0
7)	new	2	0	1	1	1	0	0	0	0	2	1	1	0	0	0	0	0	2	0	0	0	2	0	0	0	0
8)	new	1	0	2	1	1	0	1	0	0	1	1	1	0	0	0	0	0	1	1	0	0	1	1	0	0	0
9)	new	1	0	0	0	0	1	0	0	0	2	0	0	1	1	2	0	0	0	0	0	0	0	0	2	0	0
10)	new	1	0	0	0	0	1	0	0	0	2	0	0	1	1	0	0	2	0	0	0	2	0	0	0	0	0
11)	new	0	0	1	0	0	1	1	0	0	1	0	0	1	1	0	0	0	1	1	0	0	1	1	0	0	0

Memoir rows not reproduced: 2) (not feasible on the Memoir's modules (all element orders, full power maps)); 3) (not feasible on the Memoir's modules (all element orders, full power maps)); 4) (not feasible on the Memoir's modules (all element orders, full power maps)); 5) (not feasible on the Memoir's modules (all element orders, full power maps)); 6) (not feasible on the Memoir's modules (all element orders, full power maps)); 7) (not feasible on the Memoir's modules (all element orders, full power maps)); 8) (not feasible on the Memoir's modules (all element orders, full power maps)); 9) (not feasible on the Memoir's modules (all element orders, full power maps)); 10) (not feasible on the Memoir's modules (all element orders, full power maps)).

Rows marked **new** do not appear in the Memoir.

Modules tested: V133, V912, V56.

Table 6.220: $U_3(3) < E_7$, $p = 7$

		V_{133}												V_{56}											
		1	6	7_a	7_b	7_c	14	21_a	21_b	21_c	26	28_a	28_b	1	6	7_a	7_b	7_c	14	21_a	21_b	21_c	26	28_a	28_b
1)	P, N	1	2	0	0	0	0	0	1	1	3	0	0	0	0	0	0	0	0	0	0	0	0	1	1
2)	P, N	1	2	1	0	0	1	1	0	0	3	0	0	0	0	0	1	1	0	0	1	1	0	0	0
3)		2	0	2	1	1	0	1	0	0	1	1	1	0	0	0	1	1	0	0	1	1	0	0	0
4)	P, N	3	2	0	0	0	1	0	0	0	4	0	0	0	0	2	0	0	0	2	0	0	0	0	0
5)	N	3	2	0	0	0	1	0	0	0	4	0	0	4	0	0	0	0	0	0	0	0	2	0	0
6)	N	4	0	1	1	1	0	0	0	0	2	1	1	0	0	2	0	0	0	2	0	0	0	0	0
7)	N	4	0	1	1	1	0	0	0	0	2	1	1	4	0	0	0	0	0	0	0	0	2	0	0
8)	N	6	0	5	0	0	1	0	0	0	3	0	0	0	0	4	0	0	2	0	0	0	0	0	0
9)		6	6	0	2	2	3	1	0	0	0	0	0	4	4	0	0	0	2	0	0	0	0	0	0
10)		14	0	0	0	0	7	1	0	0	0	0	0	0	7	0	1	1	0	0	0	0	0	0	0
11)		21	0	14	0	0	1	0	0	0	0	0	0	14	0	6	0	0	0	0	0	0	0	0	0

Modules tested: V133, V912, V56.

Table 6.221: $U_3(8) < E_7$, $p \neq 2$

	V_{133} 133_a	V_{56} 56
1) P	1	1

Modules tested: V133, V912, V56.

Table 6.222: $U_4(2) < E_7$, $p = 0$ or $p \nmid |H|$

	V_{133}									V_{56}						
	1	5_a	5_b	6	10_a	10_b	15_a	20	24	1	5_a	5_b	6	10_a	10_b	15_a
1)	4	0	0	4	2	2	3	1	0	2	0	0	4	0	0	2
2)	8	0	0	0	0	0	7	1	0	0	0	0	6	1	1	0
3)	9	4	4	0	3	3	0	0	1	6	3	3	0	1	1	0

Modules tested: V133, V912, V56.

Table 6.223: $U_4(2) < E_7$, $p = 5$

	V_{133}									V_{56}						
	1	5_a	5_b	6	10_a	10_b	15_a	20	23	1	5_a	5_b	6	10_a	10_b	15_a
1)	4	0	0	4	2	2	3	1	0	2	0	0	4	0	0	2
2)	8	0	0	0	0	0	7	1	0	0	0	0	6	1	1	0
3)	10	4	4	0	3	3	0	0	1	6	3	3	0	1	1	0

Modules tested: V133, V912, V56.

Table 6.224: $Sp_6(2) < E_7$, $p = 0$ or $p \nmid |H|$

	V_{133}					V_{56}	
	1	7	21_a	27	35_a	7	21_a
1)	1	2	1	1	2	2	2

Modules tested: V133, V912, V56.

Table 6.225: $Sp_6(2) < E_7, p = 7$

	V_{133}					V_{56}	
	1	7	21_a	26	35_a	7	21_a
1)	2	2	1	1	2	2	2

Modules tested: V133, V912, V56.

Table 6.226: $Sp_6(2) < E_7, p = 5$

	V_{133}					V_{56}	
	1	7	21_a	27	35_a	7	21_a
1)	1	2	1	1	2	2	2

Modules tested: V133, V912, V56.

Table 6.227: $Sp_6(2) < E_7, p = 3$

	V_{133}					V_{56}	
	1	7	21	27	35	7	21
1)	1	2	1	1	2	2	2

Modules tested: V133, V912, L856, V56.

Table 6.228: $\Omega_8^+(2) < E_7, p \neq 2$

	V_{133}					V_{56}
	28	35_a	35_b	35_c	28	
1) P	1	1	1	1	2	

Modules tested: V133, V912, V56.

Table 6.229: ${}^3D_4(2) < E_7, p \neq 2, 3$

	V_{133}			V_{56}	
	1	26	52	1	26
1)	3	3	1	4	2

Modules tested: V133, V912, V56.

Table 6.230: ${}^3D_4(2) < E_7, p = 3$

		V_{133}			V_{56}	
		1	25	52	1	25
1)	N	6	3	1	6	2

Modules tested: V133, V912, L856, V56.

Table 6.231: ${}^2F_4(2)' < E_7, p \neq 2, 3, 5$

		V_{133}				V_{56}		
		1	27_a	27_b	78	1	27_a	27_b
1)		1	1	1	1	2	1	1

Modules tested: V133, V912, V56.

Table 6.232: ${}^2F_4(2)' < E_7, p = 5$

		V_{133}				V_{56}		
		1	27_a	27_b	78	1	27_a	27_b
1)	N	1	1	1	1	2	1	1

Modules tested: V133, V912, V56.

Table 6.233: ${}^2F_4(2)' < E_7, p = 3$

		V_{133}				V_{56}		
		1	27_a	27_b	77	1	27_a	27_b
1)		2	1	1	1	2	1	1

Modules tested: V133, V912, L856, V56.

Table 6.234: ${}^2B_2(8) < E_7, p = 5$

		V_{133}					V_{56}	
		14_a	14_b	35_a	35_b	35_c	14_a	14_b
1)	new	1	1	1	1	1	2	2

Memoir rows not reproduced: 1) (not feasible on the Memoir's modules (all element orders, full power maps)).
Rows marked **new** do not appear in the Memoir.

Modules tested: V133, V912, V56.

E_8 : feasible characters on $L(G) = V_{248}$

Table 6.235: $\text{Alt}_n < E_8$, $n \geq 10$, unique feasible character

Not recomputed: no character table available in GAP. See the Memoir.

Table 6.236: $\text{Alt}_{10} < E_8$, $p = 5$

		V_{248}						
		1	8	28	35_a	35_b	35_c	56
1)	P, N	1	2	3	1	0	0	2
2)		3	0	5	1	1	1	0

Modules tested: V_{248} , V_{3875} , V_{147250} .

Table 6.237: $\text{Alt}_{10} < E_8$, $p = 2$

		V_{248}							
		1	8	16	26	48	64_a	64_b	198
1)		2	0	0	0	1	0	0	1
2)	N	4	2	0	2	1	1	1	0
3)	N	4	2	1	2	2	1	0	0
4)	P, N	8	5	0	4	2	0	0	0
5)	N	16	1	1	8	0	0	0	0
6)	N	30	8	8	1	0	0	0	0

Modules tested: V_{248} , V_{3875} , L_{3626} , V_{147250} , L_{143376} .

Table 6.238: $\text{Alt}_9 < E_8$, $p = 0$ or $p \nmid |H|$

		V_{248}						
		1	8	27	28	35_a	35_b	56
1)		1	3	1	3	0	0	2
2)		3	1	1	5	1	1	0

Modules tested: V_{248} , V_{3875} , V_{147250} .

Table 6.239: $\text{Alt}_9 < E_8, p = 7$

	V_{248}						
	1	8	19	28	35_a	35_b	56
1)	1	4	1	3	0	0	2
2)	3	2	1	5	1	1	0

Modules tested: V248, V3875, V147250, L147002.

Table 6.240: $\text{Alt}_9 < E_8, p = 5$

	V_{248}						
	1	8	27	28	35_a	35_b	56
1)	1	3	1	3	0	0	2
2)	3	1	1	5	1	1	0

Memoir rows not reproduced: 1) (not compatible with V3875).

Modules tested: V248, V3875, V147250.

Table 6.241: $\text{Alt}_9 < E_8, p = 3$

	V_{248}				
	1	7	21	27	35
1) P, N	4	6	5	1	2

Modules tested: V248, V3875, V147250, L113243.

Table 6.242: $\text{Alt}_9 < E_8, p = 2$

	V_{248}								
	1	8_a	8_b	8_c	20_a	20_b	26	48	78
1) P, N	2	2	1	1	1	1	0	2	1
2) N	4	0	0	0	1	1	0	1	2
3) P, N	4	0	1	0	2	2	0	0	2
4) P, N	4	2	3	2	1	1	2	2	0
5) P, N	4	2	3	3	2	2	2	1	0
6) P, N	6	0	3	0	1	1	2	1	1
7) P, N	6	0	2	2	2	2	2	0	1
8) P, N	8	0	4	2	1	1	4	1	0
9) P, N	8	0	5	2	2	2	4	0	0

		V ₂₄₈								
		1	8 _a	8 _b	8 _c	20 _a	20 _b	26	48	78
10)	N	8	5	0	0	0	0	4	2	0
11)	N	16	1	1	1	0	0	8	0	0
12)		30	8	8	8	0	0	1	0	0

Modules tested: V248, V3875, L3626, V147250, L143376.

Table 6.243: $\text{Alt}_8 < E_8$, $p = 0$ or $p \nmid |H|$

		V ₂₄₈							
		1	7	14	20	21 _a	28	35	70
1)		3	2	1	0	0	4	1	1
2)		4	7	0	1	5	0	2	0

Memoir rows not reproduced: 1) (not compatible with V147250).

Modules tested: V248, V3875, V147250.

Table 6.244: $\text{Alt}_8 < E_8$, $p = 7$

		V ₂₄₈								
		1	7	14	19	21 _a	28	35	45	70
1)	N	3	2	1	0	0	4	1	0	1
2)		4	1	0	2	0	3	0	1	1
3)		5	7	0	1	5	0	2	0	0

Modules tested: V248, V3875, V147250, L147002.

Table 6.245: $\text{Alt}_8 < E_8$, $p = 5$

		V ₂₄₈								
		1	7	13	20	21 _a	21 _b	35	43	70
1)		3	4	0	1	0	4	0	1	1
2)		3	4	0	1	1	3	0	1	1
3)		4	6	1	0	0	4	1	0	1
4)		4	7	0	1	5	0	2	0	0

Modules tested: V248, V3875, V147250.

Table 6.246: $\text{Alt}_8 < E_8, p = 3$

		V_{248}					
		1	7	13	21	28	35
1)	N	4	3	1	0	5	2
2)	N	4	8	1	5	0	2

Modules tested: V248, V3875, V147250, L113243.

Table 6.247: $\text{Alt}_7 < E_8, p = 0$ or $p \nmid |H|$

		V_{248}								
		1	6	10_a	10_b	14_a	14_b	15	21	35
1)	P	0	0	1	1	2	0	4	0	4
2)		1	0	2	2	1	2	4	0	3
3)		2	0	3	3	0	4	4	0	2
4)		2	1	0	0	5	4	2	4	0
5)		2	1	0	0	2	7	2	4	0
6)		2	2	0	0	5	4	3	3	0
7)		2	2	0	0	2	7	3	3	0
8)		4	1	0	0	5	4	0	2	2
9)		4	1	0	0	2	7	0	2	2
10)		4	2	0	0	5	4	1	1	2
11)		4	2	0	0	2	7	1	1	2
12)		5	1	1	1	4	6	0	2	1
13)		5	2	1	1	4	6	1	1	1
14)		11	13	2	2	1	0	7	0	0

Modules tested: V248, V3875, V147250.

Table 6.248: $\text{Alt}_7 < E_8, p = 7$

		V_{248}						
		1	5	10	14_a	14_b	21	35
1)	P	0	0	8	0	4	2	2
2)	P	0	2	7	1	2	1	3
3)	P	0	4	6	2	0	0	4
4)	P, N	1	2	9	0	4	1	2
5)	P, N	1	4	8	1	2	0	3
6)	P, N	2	3	0	3	5	4	1
7)	P, N	2	3	0	6	2	4	1

		V_{248}						
		1	5	10	14_a	14_b	21	35
8)	P, N	2	4	10	0	4	0	2
9)	N	3	3	2	2	7	4	0
10)	N	3	3	2	5	4	4	0
11)	P, N	3	5	1	3	5	3	1
12)	P, N	3	5	1	6	2	3	1
13)	P, N	4	5	3	2	7	3	0
14)	P, N	4	5	3	5	4	3	0
15)		5	1	0	2	7	2	2
16)		5	1	0	5	4	2	2
17)		6	1	2	4	6	2	1
18)	N	6	3	1	2	7	1	2
19)	N	6	3	1	5	4	1	2
20)	N	7	3	3	4	6	1	1
21)	N	15	2	2	1	0	9	0
22)	N	24	20	11	1	0	0	0

Modules tested: V248, V3875, V147250, L147002.

Table 6.249: $\text{Alt}_7 < E_8$, $p = 5$

		V_{248}							
		1	6	8	10_a	10_b	13	15	35
1)	P	0	0	4	1	1	2	2	4
2)	P	0	1	3	1	1	1	3	4
3)	P	0	2	2	1	1	0	4	4
4)	P	0	4	12	4	4	1	0	1
5)		2	0	2	1	1	0	0	6
6)	N	3	0	2	2	2	3	3	3
7)	N	3	1	1	2	2	2	4	3
8)	N	3	3	11	5	5	3	0	0
9)	P, N	3	7	14	0	0	7	0	0
10)		5	0	0	2	2	1	1	5
11)	N	6	0	0	3	3	4	4	2
12)	P, N	6	6	9	0	0	8	2	0
13)	P, N	6	7	8	0	0	7	3	0
14)	N	8	6	7	0	0	6	0	2
15)	N	8	7	6	0	0	5	1	2
16)	P, N	9	3	6	0	0	11	2	0
17)	P, N	9	4	5	0	0	10	3	0
18)	N	11	3	4	0	0	9	0	2

		V_{248}							
		1	6	8	10_a	10_b	13	15	35
19)	N	11	4	3	0	0	8	1	2
20)	N	11	5	6	1	1	8	0	1
21)	N	11	6	5	1	1	7	1	1
22)		11	14	1	2	2	0	7	0
23)		28	0	25	1	1	0	0	0

Modules tested: V248, V3875, V147250.

Table 6.250: $\text{Alt}_7 < E_8$, $p = 3$

		V_{248}					
		1	6	10_a	10_b	13	15
1)	P, N	10	0	5	5	6	4
2)	P, N	11	5	0	0	9	6
3)	N	12	13	2	2	1	7
4)	P, N	17	3	2	2	11	2

Modules tested: V248, V3875, V147250, L113243.

Table 6.251: $\text{Alt}_7 < E_8$, $p = 2$

		V_{248}					
		1	4_a	4_b	6	14	20
1)	P, N	8	1	1	0	8	6
2)	P, N	8	1	1	1	9	5
3)	P, N	8	1	1	2	10	4
4)	P, N	8	4	4	4	6	5
5)	P, N	8	4	4	5	7	4
6)	P, N	8	4	4	6	8	3
7)	N	8	7	7	8	4	4
8)	N	8	7	7	9	5	3
9)	N	8	7	7	10	6	2
10)	N	18	2	2	9	0	8
11)	N	18	2	2	10	1	7
12)	N	18	2	2	17	8	0
13)		46	16	16	9	0	1
14)		46	16	16	10	1	0

Modules tested: V248, V3875, L3626, V147250, L143376.

Table 6.252: $\text{Alt}_6 < E_8$, $p = 0$ or $p \nmid |H|$

		V_{248}					
		1	5_a	5_b	8_a	8_b	10
1)	P	0	0	0	6	6	8
2)	P	0	6	0	4	4	6
3)	P	0	2	2	7	7	4
4)	P	0	3	3	4	4	6
5)		1	4	1	6	6	4
6)		1	2	2	10	5	3
7)		1	5	2	3	3	6
8)		1	7	7	6	6	9
9)		2	0	0	7	7	6
10)		2	6	0	5	5	4
11)		2	1	1	4	4	8
12)		2	4	1	9	4	3
13)		2	3	3	5	5	4
14)		2	4	4	2	2	6
15)		2	9	6	5	5	9
16)		3	0	0	10	5	5
17)		3	3	0	3	3	8
18)		3	6	0	8	3	3
19)		3	5	2	4	4	4
20)		3	3	3	8	3	3
21)		3	5	5	6	6	11
22)		3	11	5	4	4	9
23)		3	7	7	7	7	7
24)		3	8	8	4	4	9
25)		4	1	1	5	5	6
26)		4	2	2	2	2	8
27)		4	5	2	7	2	3
28)		4	4	4	3	3	4
29)		4	7	4	5	5	11
30)		4	9	6	6	6	7
31)		4	7	7	10	5	6
32)		4	10	7	3	3	9
33)		5	3	0	4	4	6
34)		5	1	1	8	3	5
35)		5	9	3	4	4	11
36)		5	4	4	6	1	3
37)		5	5	5	7	7	9
38)		5	11	5	5	5	7
39)		5	6	6	4	4	11

	1	5 _a	5 _b	V ₂₄₈		9	10
			8 _a	8 _b			
40)	5	9	6	9	4	6	1
41)	5	8	8	5	5	7	2
42)	5	9	9	2	2	9	4
43)	6	3	0	7	2	5	11
44)	6	2	2	3	3	6	12
45)	6	7	4	6	6	9	1
46)	6	5	5	10	5	8	0
47)	6	8	5	3	3	11	3
48)	6	11	5	8	3	6	2
49)	6	10	7	4	4	7	3
50)	6	8	8	8	3	6	2
51)	7	2	2	6	1	5	12
52)	7	9	3	5	5	9	2
53)	7	7	4	9	4	8	1
54)	7	6	6	5	5	9	2
55)	7	7	7	2	2	11	4
56)	7	10	7	7	2	6	3
57)	7	9	9	3	3	7	4
58)	8	0	0	20	0	0	8
59)	8	9	3	8	3	8	2
60)	8	8	5	4	4	9	3
61)	8	6	6	8	3	8	2
62)	8	9	9	6	1	6	4
63)	9	8	5	7	2	8	3
64)	9	7	7	20	0	1	0
65)	9	7	7	3	3	9	4
66)	10	7	7	6	1	8	4
67)	11	0	0	4	4	17	2
68)	11	5	5	20	0	3	0
69)	13	0	0	5	5	15	2
70)	13	1	1	2	2	17	4
71)	13	5	5	11	11	1	0
72)	14	0	0	8	3	14	2
73)	14	7	4	10	10	1	1
74)	15	1	1	3	3	15	4
75)	15	9	3	9	9	1	2
76)	15	6	6	9	9	1	2
77)	16	1	1	6	1	14	4
78)	16	8	5	8	8	1	3
79)	17	7	7	7	7	1	4

	V_{248}						
	1	5_a	5_b	8_a	8_b	9	10
80)	21	0	0	9	9	7	2
81)	23	1	1	7	7	7	4
82)	24	21	0	0	0	1	11
83)	28	0	0	25	0	0	2

Modules tested: V248, V3875, V147250.

Table 6.253: $\text{Alt}_6 < E_8$, $p = 5$

		V_{248}				
		1	5_a	5_b	8	10
1)	P, N	4	2	2	18	8
2)	P, N	5	4	1	16	9
3)	P, N	6	6	0	14	10
4)	P, N	6	3	3	14	10
5)	P, N	7	5	2	12	11
6)	P, N	8	0	0	20	8
7)	P, N	8	4	4	10	12
8)	P, N	10	1	1	16	10
9)	P, N	10	7	7	21	0
10)	P, N	11	3	0	14	11
11)	P, N	11	9	6	19	1
12)	N	12	2	2	12	12
13)	P, N	12	11	5	17	2
14)	P, N	12	8	8	17	2
15)	P, N	13	10	7	15	3
16)	P, N	14	5	5	23	0
17)	N	14	9	9	13	4
18)	P, N	15	7	4	21	1
19)	P, N	16	9	3	19	2
20)	P, N	16	6	6	19	2
21)	N	17	8	5	17	3
22)	N	18	7	7	15	4
23)		25	21	0	1	11
24)	N	28	0	0	25	2
25)	N	30	1	1	21	4

Memoir rows not reproduced: 23) (not compatible with V3875).

Modules tested: V248, V3875, V147250.

Table 6.254: $\text{Alt}_5 < E_8$, $p = 0$ or $p \nmid |H|$

		V_{248}				
		1	3_a	3_b	4	5
1)	P	0	14	14	16	20
2)		2	15	15	14	20
3)		3	18	13	13	20
4)		3	14	14	19	17
5)		5	15	15	17	17
6)		6	18	13	16	17
7)		6	14	14	22	14
8)		8	6	6	16	28
9)		8	28	8	8	20
10)		8	15	15	20	14
11)		9	18	13	19	14
12)		10	7	7	14	28
13)		10	19	19	6	20
14)		11	10	5	13	28
15)		11	6	6	19	25
16)		11	28	8	11	17
17)		13	7	7	17	25
18)		13	19	19	9	17
19)		14	10	5	16	25
20)		14	6	6	22	22
21)		14	28	8	14	14
22)		16	20	0	8	28
23)		16	7	7	20	22
24)		16	19	19	12	14
25)		17	10	5	19	22
26)		18	11	11	6	28
27)		19	20	0	11	25
28)		20	35	10	2	17
29)		21	11	11	9	25
30)		22	20	0	14	22
31)		23	35	10	5	14
32)		24	11	11	12	22
33)		28	50	0	0	14
34)		28	27	2	2	25
35)		31	27	2	5	22
36)		35	6	6	43	1
37)		37	7	7	41	1
38)		38	10	5	40	1
39)		43	20	0	35	1

		V_{248}				
		1	3_a	3_b	4	5
40)		45	11	11	33	1
41)		52	27	2	26	1
42)		78	55	0	0	1

Modules tested: V248, V3875, V147250.

Table 6.255: $\text{Alt}_5 < E_8$, $p = 3$

		V_{248}			
		1	3_a	3_b	4
1)	P, N	20	14	14	36
2)	P, N	22	15	15	34
3)	P, N	23	18	13	33
4)	N	28	28	8	28
5)	N	30	19	19	26
6)	P, N	36	6	6	44
7)	N	37	35	10	19
8)	P, N	38	7	7	42
9)	P, N	39	10	5	41
10)	N	42	50	0	14
11)	N	44	20	0	36
12)	N	46	11	11	34
13)	N	53	27	2	27
14)		79	55	0	1

Modules tested: V248, V3875, V147250, L113243.

Table 6.256: $M_{11} < E_8$, $p = 0$ or $p \nmid |H|$

	V_{248}						
	1	10_a	11	16_a	16_b	45	55
1)	9	0	6	4	4	1	0
2)	10	0	5	4	4	0	1
3)	15	6	0	4	4	1	0

Modules tested: V248, V3875, V147250.

Table 6.257: $M_{11} < E_8, p = 11$

		V_{248}							
		1	9	10_a	10_b	11	16	44	55
1)	P	0	2	3	3	0	1	1	2
2)	P, N	2	3	0	0	1	2	4	0
3)	P, N	3	4	0	0	2	0	3	1
4)	P, N	4	6	4	4	0	0	0	2
5)		9	1	1	1	6	9	0	0
6)		10	0	0	0	5	8	0	1
7)	N	21	7	1	1	0	9	0	0

Modules tested: V248, V3875, V147250.

Table 6.258: $M_{11} < E_8, p = 5$

		V_{248}						
		1	10_a	11	16_a	16_b	45	55
1)		3	1	0	0	0	4	1
2)	N	4	0	0	2	2	4	0
3)	N	7	1	7	2	2	2	0
4)	N	8	1	6	2	2	1	1
5)	N	9	0	6	4	4	1	0
6)	N	9	1	5	2	2	0	2
7)	N	10	0	5	4	4	0	1
8)	N	14	7	0	2	2	1	1
9)	N	15	6	0	4	4	1	0

Modules tested: V248, V3875, V147250.

Table 6.259: $M_{11} < E_8, p = 3$

		V_{248}							
		1	5_a	5_b	10_a	10_b	10_c	24	45
1)	P	0	0	0	2	0	0	2	4
2)	P, N	1	2	2	0	1	1	3	3
3)	P, N	2	2	2	0	2	2	4	2
4)	P, N	4	5	5	5	0	0	6	0
5)	P, N	4	7	7	3	0	0	6	0
6)	N	23	4	4	6	4	4	0	1
7)	N	23	10	10	0	4	4	0	1

		V_{248}							
		1	5_a	5_b	10_a	10_b	10_c	24	45
8)	N	24	10	10	0	5	5	1	0

Modules tested: V248, V3875, V147250, L113243.

Table 6.260: $M_{11} < E_8$, $p = 2$

		V_{248}				
		1	10	16_a	16_b	44
1)	N	14	3	5	5	1
2)	N	16	6	4	4	1
3)	N	18	9	3	3	1

Modules tested: V248, V3875, L3626, V147250, L143376.

Table 6.261: $M_{12} < E_8$, $p = 5$

		V_{248}				
		1	11_a	16_a	16_b	78
1)		8	6	3	3	1

Modules tested: V248, V3875, V147250.

Table 6.262: $M_{12} < E_8$, $p = 2$

		V_{248}				
		1	10	16_a	16_b	44
1)	N	16	6	4	4	1

Modules tested: V248, V3875, L3626, V147250, L143376.

Table 6.263: $J_1 < E_8$, $p = 11$

		V_{248}					
		1	7	14	27	64	77_c
1)		1	0	3	0	2	1
2)		6	13	5	3	0	0
3)		8	0	1	6	1	0

	V_{248}					
	1	7	14	27	64	77_c
4)	52	26	1	0	0	0

Modules tested: V248, V3875, V147250.

Table 6.264: $J_2 < E_8$, $p = 2$

		V_{248}							
		1	6_a	6_b	14_a	14_b	36	64_b	84
1)		2	1	0	1	3	1	1	1
2)	N	4	2	2	4	0	1	2	0
3)	N	8	4	3	2	1	2	0	1
4)	N	14	6	6	7	0	0	1	0
5)	N	16	8	8	1	1	3	0	0
6)	N	22	16	3	8	0	0	0	0
7)	N	78	26	0	1	0	0	0	0

Modules tested: V248, V3875, L3626, V147250, L143376.

Table 6.265: $J_3 < E_8$, $p = 2$

		V_{248}		
		80	84_a	84_b
1)	P	1	1	1

Modules tested: V248, V3875, L3626, V147250, L143376.

Table 6.266: $Th < E_8$, $p = 3$ (irreducible on V_{248})

Not recomputed: no Brauer table available in GAP. See the Memoir.

Table 6.267: $L_2(7) < E_8$, $p = 0$ or $p \nmid |H|$

		V_{248}					
		1	3_a	3_b	6	7	8
1)	P	0	5	5	6	10	14
2)		2	7	7	6	8	14
3)		3	0	0	14	7	14

	V_{248}					
	1	3_a	3_b	6	7	8
4)	3	5	5	6	13	11
5)	5	2	2	14	5	14
6)	5	7	7	6	11	11
7)	6	0	0	14	10	11
8)	6	5	5	6	16	8
9)	8	2	2	14	8	11
10)	8	7	7	6	14	8
11)	9	0	0	14	13	8
12)	11	2	2	14	11	8
13)	14	8	8	14	2	11
14)	17	8	8	14	5	8
15)	22	21	21	6	0	8
16)	28	1	1	0	2	25
17)	31	1	1	0	5	22
18)	52	1	1	0	26	1
19)	78	27	27	0	0	1

Modules tested: V248, V3875, V147250.

Table 6.268: $L_2(7) < E_8$, $p = 3$

		V_{248}				
		1	3_a	3_b	6	7
1)	P, N	14	5	5	6	24
2)	P, N	16	7	7	6	22
3)	P, N	17	0	0	14	21
4)	N	19	2	2	14	19
5)	N	25	8	8	14	13
6)	N	30	21	21	6	8
7)	N	53	1	1	0	27
8)		79	27	27	0	1

Modules tested: V248, V3875, V147250, L113243.

Table 6.269: $L_2(8) < E_8$, $p = 0$ or $p \nmid |H|$

		V ₂₄₈								
		1	7 _a	7 _b	7 _c	7 _d	8	9 _a	9 _b	9 _c
1)	P P	0	3	5	5	5	4	4	3	3
2)		0	6	5	4	3	4	4	3	3
3)		1	3	5	5	5	5	3	3	3
4)		1	4	5	5	5	3	4	3	3
5)		1	6	5	4	3	5	3	3	3
6)		2	2	8	6	4	2	4	3	3
7)		2	2	7	5	6	2	4	3	3
8)		2	3	5	5	5	6	5	2	1
9)		2	4	5	5	5	4	3	3	3
10)		2	5	8	5	2	2	4	3	3
11)		2	6	5	4	3	6	5	2	1
12)		3	2	8	6	4	3	3	3	3
13)		3	2	7	5	6	3	3	3	3
14)		3	4	5	5	5	5	5	2	1
15)		3	5	8	5	2	3	3	3	3
16)		3	6	5	5	5	1	4	3	3
17)		4	2	8	6	4	4	5	2	1
18)		4	2	7	5	6	4	5	2	1
19)		4	5	8	5	2	4	5	2	1
20)		4	6	5	5	5	2	3	3	3
21)		4	7	11	2	2	0	4	3	3
22)		5	0	1	1	1	3	8	7	7
23)		5	6	5	5	5	3	5	2	1
24)		5	7	11	2	2	1	3	3	3
25)		6	0	1	1	1	4	7	7	7
26)		6	5	13	5	0	0	3	3	3
27)		6	5	9	4	5	0	3	3	3
28)		6	7	11	2	2	2	5	2	1
29)		7	0	1	1	1	5	12	2	6
30)		7	0	1	1	1	5	9	6	5
31)		7	2	1	1	1	1	8	7	7
32)		7	5	13	5	0	1	5	2	1
33)		7	5	9	4	5	1	5	2	1
34)		8	2	1	1	1	2	7	7	7
35)		8	12	11	1	0	0	5	2	1
36)		9	2	1	1	1	3	12	2	6
37)		9	2	1	1	1	3	9	6	5
38)		10	1	5	0	1	0	7	7	7
39)		11	1	5	0	1	1	12	2	6

	V_{248}								
	1	7_a	7_b	7_c	7_d	8	9_a	9_b	9_c
40)	11	1	5	0	1	1	9	6	5
41)	12	0	1	1	1	10	8	0	7
42)	12	0	1	1	1	10	14	0	1
43)	14	2	1	1	1	8	8	0	7
44)	14	2	1	1	1	8	14	0	1
45)	16	1	5	0	1	6	8	0	7
46)	16	1	5	0	1	6	14	0	1
47)	21	9	1	1	1	1	8	0	7
48)	21	9	1	1	1	1	14	0	1
49)	27	0	1	1	1	25	0	0	0
50)	29	2	1	1	1	23	0	0	0
51)	31	1	5	0	1	21	0	0	0
52)	36	9	1	1	1	16	0	0	0
53)	52	1	26	1	0	0	0	0	0

Modules tested: V248, V3875, V147250.

Table 6.270: $L_2(8) < E_8$, $p = 7$

	V_{248}					
	1	7_a	7_b	7_c	7_d	8
1) P, N	10	3	5	5	5	14
2) P, N	10	6	5	4	3	14
3) P, N	11	4	5	5	5	13
4) N	12	2	8	6	4	12
5) N	12	2	7	5	6	12
6) N	12	5	8	5	2	12
7) N	13	6	5	5	5	11
8) N	14	7	11	2	2	10
9) N	15	5	13	5	0	9
10) N	15	5	9	4	5	9
11) N	16	12	11	1	0	8
12) N	20	13	5	5	5	4
13) N	27	0	1	1	1	25
14) N	29	2	1	1	1	23
15) N	31	1	5	0	1	21
16) N	36	9	1	1	1	16
17)	52	1	26	1	0	0

Modules tested: V248, V3875, V147250, L147002.

Table 6.271: $L_2(8) < E_8$, $p = 3$

		V_{248}				
		1	7	9_a	9_b	9_c
1)	P, N	4	22	4	3	3
2)	P, N	6	23	3	3	3
3)	N	8	6	8	7	7
4)	P, N	8	24	5	2	1
5)	N	10	7	7	7	7
6)	N	12	8	12	2	6
7)	N	12	8	9	6	5
8)	N	22	13	14	0	1
9)	N	22	13	8	0	7
10)	N	52	28	0	0	0

Modules tested: V248, V3875, V147250, L113243.

Table 6.272: $L_2(11) < E_8$, $p = 0$ or $p \nmid |H|$

		V_{248}							
		1	5_a	5_b	10_a	10_b	11	12_a	12_b
1)	P	0	2	2	4	0	4	6	6
2)		1	0	0	5	2	3	6	6
3)		2	2	2	4	0	6	5	5
4)		3	0	0	5	2	5	5	5
5)		3	2	2	4	0	7	7	2
6)		4	0	0	5	2	6	7	2
7)		7	1	1	9	1	1	5	5
8)		7	1	1	1	9	1	5	5
9)		7	4	4	1	6	1	5	5
10)		8	1	1	9	1	2	7	2
11)		8	1	1	1	9	2	7	2
12)		8	2	2	2	8	0	5	5
13)		8	4	4	1	6	2	7	2
14)		8	8	8	2	2	0	5	5
15)		9	2	2	2	8	1	7	2
16)		9	8	8	2	2	1	7	2
17)		10	2	2	4	0	14	1	1
18)		11	0	0	5	2	13	1	1
19)		15	1	1	9	1	9	1	1

	V ₂₄₈							
	1	5 _a	5 _b	10 _a	10 _b	11	12 _a	12 _b
20)	15	1	1	1	9	9	1	1
21)	15	4	4	1	6	9	1	1
22)	16	2	2	2	8	8	1	1
23)	16	8	8	2	2	8	1	1
24)	24	10	10	10	0	0	1	1

Modules tested: V248, V3875, V147250.

Table 6.273: $L_2(11) < E_8$, $p = 5$

	V ₂₄₈					
	1	5 _a	5 _b	10 _a	10 _b	11
1) P, N	12	2	2	4	0	16
2) P, N	13	0	0	5	2	15
3) N	17	1	1	9	1	11
4) N	17	1	1	1	9	11
5) N	17	4	4	1	6	11
6) N	18	2	2	2	8	10
7) N	18	8	8	2	2	10
8) N	26	10	10	10	0	2

Modules tested: V248, V3875, V147250.

Table 6.274: $L_2(11) < E_8$, $p = 3$

	V ₂₄₈					
	1	5 _a	5 _b	10	12 _a	12 _b
1) P, N	4	2	2	8	6	6
2)	4	10	10	0	6	6
3) P, N	8	2	2	10	5	5
4) N	8	10	10	2	5	5
5) P, N	10	2	2	11	7	2
6) N	10	10	10	3	7	2
7) N	24	2	2	18	1	1
8) N	24	10	10	10	1	1

Modules tested: V248, V3875, V147250, L113243.

Table 6.275: $L_2(11) < E_8, p = 2$

		V_{248}					
		1	5_a	5_b	10	12_a	12_b
1)		4	0	0	10	6	6
2)	P, N	4	3	3	7	6	6
3)	P, N	4	6	6	4	6	6
4)	N	8	2	2	10	5	5
5)	P, N	8	5	5	7	5	5
6)	P, N	8	8	8	4	5	5
7)	N	10	3	3	10	7	2
8)	P, N	10	6	6	7	7	2
9)	P, N	10	9	9	4	7	2
10)	N	24	10	10	10	1	1
11)	P, N	24	13	13	7	1	1
12)	P, N	24	16	16	4	1	1

Modules tested: V248, V3875, L3626, V147250, L143376.

Table 6.276: $L_2(13) < E_8, p = 0$ or $p \nmid |H|$

		V_{248}								
		1	7_a	7_b	12_a	12_b	12_c	13	14_a	14_b
1)	P	0	1	1	3	3	2	2	2	6
2)		1	1	1	3	3	3	1	2	6
3)		1	3	3	3	3	2	3	1	4
4)		2	1	1	5	1	4	0	2	6
5)		2	2	2	1	1	0	2	4	8
6)		2	3	3	3	3	3	2	1	4
7)		2	5	5	3	3	2	4	0	2
8)		3	2	2	1	1	1	1	4	8
9)		3	3	3	5	1	4	1	1	4
10)		3	4	4	1	1	0	3	3	6
11)		3	5	5	3	3	3	3	0	2
12)		4	0	0	1	1	0	4	10	2
13)		4	4	4	1	1	1	2	3	6
14)		4	5	5	5	1	4	2	0	2
15)		4	6	6	1	1	0	4	2	4
16)		5	0	0	1	1	1	3	10	2
17)		5	2	2	1	1	0	5	9	0
18)		5	6	6	1	1	1	3	2	4
19)		6	13	0	0	0	0	3	3	5

	V_{248}								
	1	7_a	7_b	12_a	12_b	12_c	13	14_a	14_b
20)	6	2	2	1	1	1	4	9	0
21)	7	0	0	1	1	0	7	7	2
22)	8	0	0	1	1	1	6	7	2
23)	8	2	2	1	1	0	8	6	0
24)	9	2	2	1	1	1	7	6	0
25)	14	0	0	8	1	0	0	7	2
26)	15	2	2	8	1	0	1	6	0
27)	52	26	0	0	0	0	0	0	1

Modules tested: V248, V3875, V147250.

Table 6.277: $L_2(13) < E_8, p = 7$

		V_{248}					
		1	7_a	7_b	12	14_a	14_b
1)	P, N	2	1	1	10	2	6
2)	N	4	2	2	4	4	8
3)	P, N	4	3	3	11	1	4
4)	N	6	4	4	5	3	6
5)	P, N	6	5	5	12	0	2
6)	N	8	0	0	6	10	2
7)	N	8	6	6	6	2	4
8)	N	9	13	0	3	3	5
9)	N	10	2	2	7	9	0
10)	N	14	0	0	9	7	2
11)	N	16	2	2	10	6	0
12)		52	26	0	0	0	1

Modules tested: V248, V3875, V147250, L147002.

Table 6.278: $L_2(13) < E_8, p = 3$

		V_{248}						
		1	7_a	7_b	12_a	12_b	12_c	13
1)	P, N	2	7	7	3	3	2	4
2)	N	3	7	7	3	3	3	3
3)	N	4	7	7	5	1	4	2
4)	N	6	10	10	1	1	0	6
5)	N	7	10	10	1	1	1	5

		V_{248}						
		1	7_a	7_b	12_a	12_b	12_c	13
6)	N	9	18	5	0	0	0	6
7)	N	14	2	2	1	1	0	14
8)	N	15	2	2	1	1	1	13
9)	N	21	2	2	8	1	0	7
10)		52	27	1	0	0	0	0

Modules tested: V248, V3875, V147250, L113243.

Table 6.279: $L_2(13) < E_8$, $p = 2$

		V_{248}						
		1	6_a	6_b	12_a	12_b	12_c	14
1)	N P, N	4	1	1	5	1	4	8
2)		4	2	2	3	3	3	8
3)		4	3	3	3	3	2	8
4)	N	8	3	3	1	1	1	12
5)	N	8	4	4	1	1	0	12
6)	N	10	4	4	5	1	4	5
7)	N	10	5	5	3	3	3	5
8)	P, N	10	6	6	3	3	2	5
9)	N P, N	14	0	0	8	1	0	9
10)		14	6	6	1	1	1	9
11)		14	7	7	1	1	0	9
12)	N	16	13	0	0	0	0	11
13)	N	16	2	2	10	3	2	2
14)	N	16	7	7	5	1	4	2
15)	N	16	8	8	3	3	3	2
16)	P, N	16	9	9	3	3	2	2
17)	N	20	3	3	8	1	0	6
18)	N	20	9	9	1	1	1	6
19)	N	20	10	10	1	1	0	6
20)	N	22	16	3	0	0	0	8
21)	N	78	26	0	0	0	0	1

Modules tested: V248, V3875, L3626, V147250, L143376.

Table 6.280: $L_2(16) < E_8$, $p = 0$ or $p \nmid |H|$

		V_{248}																
		1	15_a	15_b	15_c	15_d	15_e	15_f	15_g	15_h	16	17_a	17_b	17_c	17_d	17_e	17_f	17_g
1)	P	0	3	1	1	0	2	2	2	1	0	0	0	0	1	1	1	1
2)	P	0	2	2	2	2	1	1	1	1	0	0	0	0	1	1	1	1
3)	P	0	2	2	1	1	2	1	1	2	0	0	0	0	1	1	1	1
4)		1	3	1	1	0	2	2	2	1	1	1	1	1	0	0	0	0
5)		1	2	2	2	2	1	1	1	1	1	1	1	1	0	0	0	0
6)		1	2	2	1	1	2	1	1	2	1	1	1	1	0	0	0	0
7)		2	4	3	1	0	0	1	3	1	0	1	1	1	0	0	0	0
8)		2	3	1	1	2	2	0	3	1	0	1	1	1	0	0	0	0

Modules tested: V248, V3875, V147250.

Table 6.281: $L_2(16) < E_8$, $p = 17$

		V_{248}								
		1	15	17_a	17_b	17_c	17_d	17_e	17_f	17_g
1)	P	0	12	0	0	0	1	1	1	1
2)	P	2	13	1	1	1	0	0	0	0
3)		10	0	0	3	3	4	0	4	0
4)		10	0	1	6	0	4	0	0	3
5)		10	0	1	2	1	4	4	1	1
6)		10	0	8	0	0	3	0	2	1
7)		10	0	8	2	1	2	0	0	1
8)		10	0	8	3	3	0	0	0	0
9)		12	1	7	1	1	1	1	1	1

Modules tested: V248, V3875, V147250.

Table 6.282: $L_2(16) < E_8$, $p = 5$

		V_{248}										
		1	15_a	15_b	15_c	15_d	15_e	15_f	15_g	15_h	16	17
1)	P	0	2	2	2	2	1	1	1	1	0	4
2)	P	0	2	2	1	1	2	1	1	2	0	4
3)	P	0	3	1	1	0	2	2	2	1	0	4
4)		3	2	2	2	2	1	1	1	1	3	1
5)		3	2	2	1	1	2	1	1	2	3	1
6)		3	3	1	1	0	2	2	2	1	3	1

	V_{248}										
	1	15_a	15_b	15_c	15_d	15_e	15_f	15_g	15_h	16	17
7)	4	3	1	1	2	2	0	3	1	2	1
8)	4	4	3	1	0	0	1	3	1	2	1
9)	5	4	4	0	0	2	2	1	1	1	1
10)	5	5	1	3	0	3	1	0	1	1	1
11)	5	3	2	2	0	2	2	3	0	1	1

Modules tested: V248, V3875, V147250.

Table 6.283: $L_2(16) < E_8$, $p = 3$

		V ₂₄₈											
		1	15 _a	15 _b	15 _c	15 _d	15 _e	15 _f	15 _g	15 _h	16	17 _a	17 _b
1)	P	0	3	1	1	0	2	2	2	1	0	2	2
2)	P	0	2	2	2	2	1	1	1	1	0	2	2
3)	P	0	2	2	1	1	2	1	1	2	0	2	2
4)		2	3	1	1	0	2	2	2	1	2	1	1
5)		2	2	2	2	2	1	1	1	1	2	1	1
6)		2	2	2	1	1	2	1	1	2	2	1	1
7)		3	4	3	1	0	0	1	3	1	1	1	1
8)		3	3	1	1	2	2	0	3	1	1	1	1
9)		4	3	2	2	0	2	2	3	0	0	1	1
10)		4	5	1	3	0	3	1	0	1	0	1	1
11)		4	4	4	0	0	2	2	1	1	0	1	1

Modules tested: V248, V3875, V147250, L113243.

Table 6.284: $L_2(17) < E_8$, $p = 0$ or $p \nmid |H|$

		V_{248}										
		1	9_a	9_b	16_a	16_b	16_c	16_d	17	18_a	18_b	18_c
1)	P	0	1	0	0	2	2	2	1	1	3	3
2)	P	0	1	0	3	2	1	0	1	1	3	3
3)		1	1	0	1	2	2	2	0	1	3	3
4)		7	8	1	1	0	0	0	0	6	1	1
5)		8	7	0	1	0	0	0	1	6	1	1
6)		10	11	4	1	0	0	0	3	0	1	1
7)		14	7	0	1	0	0	0	7	0	1	1
8)		21	7	0	8	0	0	0	0	0	1	1

Modules tested: V248, V3875, V147250.

Table 6.285: $L_2(17) < E_8, p = 3$

		V_{248}						
		1	9_a	9_b	16	18_a	18_b	18_c
1)	P, N	1	1	0	7	1	3	3
2)		7	8	1	1	6	1	1
3)	N	9	7	0	2	6	1	1
4)	N	13	11	4	4	0	1	1
5)	N	21	7	0	8	0	1	1

Modules tested: V248, V3875, V147250, L113243.

Table 6.286: $L_2(17) < E_8, p = 2$

		V_{248}						
		1	8_a	8_b	16_a	16_b	16_c	16_d
1)	N	16	4	3	2	6	1	2
2)	N	16	6	5	3	2	2	2
3)	N	16	8	7	1	2	2	2
4)	P, N	16	9	8	0	2	2	2
5)	P, N	16	9	8	3	2	1	0
6)	N	32	9	2	8	0	0	0
7)	N	32	16	9	1	0	0	0

Modules tested: V248, V3875, L3626, V147250, L143376.

Table 6.287: $L_2(19) < E_8, p = 0$ or $p \nmid |H|$

		V_{248}										
		1	9_a	9_b	18_a	18_b	18_d	19	20_a	20_b	20_c	20_d
1)	P	0	1	1	2	2	0	2	0	3	1	2
2)	P	0	1	1	2	2	0	2	3	1	1	1
3)		1	0	0	3	2	1	1	0	3	1	2
4)		1	0	0	3	2	1	1	3	1	1	1
5)		1	1	1	2	2	0	3	2	1	1	1
6)		2	0	0	3	2	1	2	2	1	1	1
7)		3	1	1	2	2	0	5	0	1	1	1
8)		4	0	0	3	2	1	4	0	1	1	1

	V_{248}											
	1	9_a	9_b	18_a	18_b	18_d	19	20_a	20_b	20_c	20_d	
9)	8	3	3	1	0	6	0	0	1	1	1	

Modules tested: V248, V3875, V147250.

Table 6.288: $L_2(19) < E_8$, $p = 5$

		V_{248}							
		1	9_a	9_b	18	20_a	20_b	20_c	20_d
1)	P, N	2	1	1	6	0	3	1	2
2)	P, N	2	1	1	6	3	1	1	1
3)	P, N	4	1	1	7	2	1	1	1
4)	P, N	8	1	1	9	0	1	1	1
5)	N	8	9	9	1	0	1	1	1

Modules tested: V248, V3875, V147250.

Table 6.289: $L_2(19) < E_8$, $p = 3$

		V_{248}						
		1	9_a	9_b	18_a	18_b	18_d	19
1)	P, N	6	1	1	2	2	0	8
2)	N	7	0	0	3	2	1	7
3)	N	11	3	3	1	0	6	3

Modules tested: V248, V3875, V147250, L113243.

Table 6.290: $L_2(19) < E_8$, $p = 2$

		V_{248}								
		1	9_a	9_b	18_a	18_b	20_a	20_b	20_c	20_d
1)		2	0	0	6	1	0	3	1	2
2)		2	0	0	6	1	3	1	1	1
3)	N	2	1	1	3	3	0	3	1	2
4)	N	2	1	1	3	3	3	1	1	1
5)	P, N	2	3	3	2	2	0	3	1	2
6)	P, N	2	3	3	2	2	3	1	1	1
7)	N	4	1	1	6	1	2	1	1	1
8)	N	4	2	2	3	3	2	1	1	1

		V_{248}								
		1	9_a	9_b	18_a	18_b	20_a	20_b	20_c	20_d
9)	P, N	4	4	4	2	2	2	1	1	1
10)	N	8	3	3	6	1	0	1	1	1
11)	N	8	4	4	3	3	0	1	1	1
12)	P, N	8	6	6	2	2	0	1	1	1

Modules tested: V248, V3875, L3626, V147250, L143376.

Table 6.291: $L_2(25) < E_8$, $p = 0$ or $p \nmid |H|$

		V_{248}												
		1	24_a	24_b	24_c	24_d	24_e	24_f	25	26_a	26_b	26_c	26_d	26_e
1)	P	0	2	1	1	0	1	1	0	0	0	0	2	2
2)	P	0	1	1	1	1	1	1	0	0	0	0	2	2
3)		14	0	0	0	0	0	0	0	7	0	0	1	1
4)		15	0	0	0	0	0	0	1	6	2	0	0	0
5)		15	0	0	0	0	0	0	1	0	2	6	0	0

Modules tested: V248, V3875, V147250.

Table 6.292: $L_2(25) < E_8$, $p = 13$

		V_{248}						
		1	24	26_a	26_b	26_c	26_d	26_e
1)	P	0	6	0	0	0	2	2
2)		14	0	7	0	0	1	1
3)		16	1	6	2	0	0	0
4)		16	1	0	2	6	0	0

Modules tested: V248, V3875, V147250.

Table 6.293: $L_2(25) < E_8$, $p = 3$

		V_{248}										
		1	13_a	13_b	24_a	24_b	24_c	24_d	24_e	24_f	25	26
1)	P	0	0	0	2	1	1	0	1	1	0	4
2)		15	6	6	0	0	0	0	0	0	1	2
3)	N	21	0	0	0	0	0	0	0	0	7	2
4)	new	0	0	0	1	1	1	1	1	1	0	4

Rows marked **new** do not appear in the Memoir.
Modules tested: V248, V3875, V147250, L113243.

Table 6.294: $L_2(25) < E_8$, $p = 2$

		V_{248}									
		1	12_a	12_b	24_a	24_b	24_c	24_d	24_e	24_f	26
1)	P	0	0	0	2	1	1	0	1	1	4
2)	P	0	0	0	1	1	1	1	1	1	4
3)	N	6	2	2	2	2	2	0	1	0	1
4)	N	6	3	3	1	1	1	1	1	1	1
5)	N	6	3	3	2	1	1	0	1	1	1
6)	N	10	7	2	0	0	0	0	0	0	5
7)		14	0	0	0	0	0	0	0	0	9
8)	N	16	10	5	0	0	0	0	0	0	2
9)	N	20	3	3	0	0	0	0	0	0	6

Modules tested: V248, V3875, L3626, V147250, L143376.

Table 6.295: $L_2(27) < E_8$, $p = 0, 13$ or $p \nmid |H|$

		V_{248}			
		1	26_a	26_b	26_e
1)		14	1	1	7

Modules tested: V248, V3875, V147250.

Table 6.296: $L_2(27) < E_8$, $p = 7$

		V_{248}			
		1	13_a	13_b	26
1)	N	14	7	7	2

Modules tested: V248, V3875, V147250, L147002.

Table 6.297: $L_2(27) < E_8$, $p = 2$

		V_{248}											
		1	13 _a	13 _b	26 _a	26 _b	26 _c	28 _a	28 _b	28 _c	28 _d	28 _e	28 _f
1)		2	0	0	1	1	1	2	1	1	1	0	1
2)		2	0	0	1	1	1	1	1	1	1	1	1
3)	N	2	1	1	1	1	0	2	1	1	1	0	1
4)	N	2	1	1	1	1	0	1	1	1	1	1	1
5)	N	4	1	1	1	1	1	4	0	0	0	0	1
6)	N	4	1	1	1	1	1	2	2	0	1	0	0
7)	N	4	2	2	1	1	0	4	0	0	0	0	1
8)	N	4	2	2	1	1	0	2	2	0	1	0	0
9)		14	0	0	8	0	1	0	0	0	0	0	0
10)	N	14	6	6	1	1	1	0	0	0	0	0	0
11)	N	14	7	7	1	1	0	0	0	0	0	0	0

Modules tested: V248, V3875, L3626, V147250, L143376.

Table 6.298: $L_2(29) < E_8$, $p = 0$ or $p \nmid |H|$

		V_{248}													
		1	15 _a	15 _b	28 _a	28 _b	28 _c	28 _d	28 _e	28 _f	28 _g	29	30 _d	30 _e	30 _f
1)	P	0	1	0	1	1	1	0	0	0	0	1	2	1	1
2)	P	0	2	1	1	1	1	0	0	0	0	1	1	1	1
3)	new	1	2	1	0	0	0	1	1	1	1	0	1	1	1
4)	new	1	1	0	0	0	0	1	1	1	1	0	2	1	1

Memoir rows read with the published correction: one constituent 28 replaced by 29 in each row.

Rows marked **new** do not appear in the Memoir.

Modules tested: V248, V3875, V147250.

Table 6.299: $L_2(29) < E_8$, $p = 7$

		V_{248}										
		1	15_a	15_b	28_a	28_b	28_c	28_d	28_e	28_f	28_g	29
1)	P	0	5	4	1	1	1	0	0	0	0	1
2)		1	5	4	0	0	0	1	1	1	1	0

Modules tested: V248, V3875, V147250, L147002.

Table 6.300: $L_2(29) < E_8$, $p = 5$

		V_{248}							
		1	15_a	15_b	28_a	28_b	30_d	30_e	30_f
1)		1	1	0	4	0	2	1	1
2)	P	1	1	0	1	3	2	1	1
3)		1	2	1	4	0	1	1	1
4)	P	1	2	1	1	3	1	1	1

Memoir rows not reproduced: 3) (not feasible on the Memoir's modules (all element orders, full power maps)); 4) (not feasible on the Memoir's modules (all element orders, full power maps)).

Modules tested: V248, V3875, V147250.

Table 6.301: $L_2(29) < E_8$, $p = 3$

		V_{248}								
		1	15_a	15_b	28_a	28_b	28_c	30_d	30_e	30_f
1)		1	1	0	0	2	2	2	1	1
2)	P, N	1	1	0	2	1	1	2	1	1
3)		1	2	1	0	2	2	1	1	1
4)	P, N	1	2	1	2	1	1	1	1	1

Modules tested: V248, V3875, V147250, L113243.

Table 6.302: $L_2(29) < E_8$, $p = 2$

		V_{248}												
		1	14_a	14_b	28_a	28_b	28_c	28_d	28_e	28_f	28_g	30_a	30_b	30_c
1)	P, N	2	1	0	0	0	0	1	1	1	1	2	1	1
2)		2	2	1	1	1	1	0	0	0	0	2	1	1
3)		4	1	0	0	1	0	2	0	2	0	1	1	1
4)		4	1	0	1	1	0	1	2	0	0	1	1	1
5)	N	4	2	1	0	0	0	1	1	1	1	1	1	1
6)	P	4	3	2	1	1	1	0	0	0	0	1	1	1

Modules tested: V248, V3875, L3626, V147250, L143376.

Table 6.303: $L_2(31) < E_8$, $p = 0$ or $p \nmid |H|$

		V_{248}													
		1	15_a	15_b	30_a	30_b	30_c	31	32_a	32_b	32_c	32_d	32_e	32_f	32_g
1)	P	0	0	0	0	2	2	0	0	0	0	1	1	1	1
2)	P	0	0	0	2	1	1	0	0	0	0	1	1	1	1
3)	P	0	1	1	1	2	0	0	0	0	0	1	1	1	1
4)	P	0	1	1	1	1	1	0	0	0	0	1	1	1	1
5)		1	0	0	0	2	2	1	1	1	1	0	0	0	0
6)		1	0	0	2	1	1	1	1	1	1	0	0	0	0
7)		1	1	1	1	2	0	1	1	1	1	0	0	0	0
8)		1	1	1	1	1	1	1	1	1	1	0	0	0	0

Modules tested: V248, V3875, V147250.

Table 6.304: $L_2(31) < E_8$, $p = 5$

		V_{248}							
		1	15 _a	15 _b	30 _a	30 _b	30 _c	31	32
1)	P	0	0	0	0	2	2	0	4
2)	P	0	0	0	2	1	1	0	4
3)	P	0	1	1	1	2	0	0	4
4)	P	0	1	1	1	1	1	0	4
5)		3	0	0	0	2	2	3	1
6)		3	0	0	2	1	1	3	1
7)		3	1	1	1	2	0	3	1
8)		3	1	1	1	1	1	3	1
9)		5	2	2	2	1	1	1	1
10)		5	3	3	1	1	1	1	1

Memoir rows not reproduced: 13) (not compatible with V3875).

Modules tested: V248, V3875, V147250.

Table 6.305: $L_2(31) < E_8$, $p = 3$

		V_{248}								
		1	15_a	15_b	30_a	30_b	30_c	31	32_a	32_b
1)	P	0	0	0	0	2	2	0	2	2
2)	P	0	0	0	2	1	1	0	2	2
3)	P	0	1	1	1	2	0	0	2	2
4)	P	0	1	1	1	1	1	0	2	2

	V_{248}								
	1	15_a	15_b	30_a	30_b	30_c	31	32_a	32_b
5)	2	0	0	0	2	2	2	1	1
6)	2	0	0	2	1	1	2	1	1
7)	2	1	1	1	2	0	2	1	1
8)	2	1	1	1	1	1	2	1	1
9)	4	2	2	2	1	1	0	1	1
10)	4	3	3	1	1	1	0	1	1

Modules tested: V248, V3875, V147250, L113243.

Table 6.306: $L_2(31) < E_8$, $p = 2$

		V_{248}									
		1	15_a	15_b	32_a	32_b	32_c	32_d	32_e	32_f	32_g
1)	P	0	4	4	0	0	0	1	1	1	1
2)	P	2	5	5	1	1	1	0	0	0	0

Modules tested: V248, V3875, L3626, V147250, L143376.

Table 6.307: $L_2(32) < E_8$, $p = 0$, 31 or $p \nmid |H|$

		V_{248}							
		31_b	31_c	31_d	31_e	31_g	31_i	31_l	31_n
1)	P	1	1	1	1	1	1	1	1

Modules tested: V248, V3875, V147250.

Table 6.308: $L_2(32) < E_8$, $p = 11$

		V_{248}	
		31_a	31_b
1)	P	7	1
2)	P	4	4

Modules tested: V248, V3875, V147250.

Table 6.309: $L_2(32) < E_8, p = 3$

		V_{248}				
		31_b	31_c	31_d	31_e	31_f
1)	P	3	1	3	1	0
2)	P	2	2	2	1	1

Modules tested: V248, V3875, V147250, L113243.

Table 6.310: $L_2(37) < E_8, p = 2$

Not recomputed: no Brauer table available in GAP. See the Memoir.

Table 6.311: $L_2(41) < E_8, p = 0, 3$ or $p \nmid |H|$

		V_{248}					
		40_c	40_f	42_a	42_c	42_g	42_i
1)	P	1	1	1	1	1	1

Modules not used (elements of large order): V3875, V147250.

Modules tested: V248.

Table 6.312: $L_2(41) < E_8, p = 7$

		V_{248}				
		40_b	42_a	42_c	42_g	42_i
1)	P	2	1	1	1	1

Modules not used (elements of large order): V3875, V147250, L147002.

Modules tested: V248.

Table 6.313: $L_2(41) < E_8, p = 5$

		V_{248}		
		40_c	40_f	42
1)	P	1	1	4

Modules not used (elements of large order): V3875, V147250.

Modules tested: V248.

Table 6.314: $L_2(41) < E_8$, $p = 2$

Not recomputed: no Brauer table available in GAP. See the Memoir.

Table 6.315: $L_2(49) < E_8$, $p = 0$ or $p \nmid |H|$

		V_{248}				
		48_a	50_h	50_i	50_j	50_k
1)	P	1	1	1	1	1

Modules tested: V248, V3875, V147250.

Table 6.316: $L_2(49) < E_8$, $p = 5$

		V_{248}				
		48	50_h	50_i	50_j	50_k
1)	P	1	1	1	1	1

Modules tested: V248, V3875, V147250.

Table 6.317: $L_2(49) < E_8$, $p = 3$

		V_{248}		
		48_a	50_b	50_c
1)	P	1	2	2

Modules tested: V248, V3875, V147250, L113243.

Table 6.318: $L_2(49) < E_8$, $p = 2$

		V_{248}					
		1	24_a	24_b	48_a	48_b	50
1)	P	0	0	0	1	0	4
2)		6	1	1	2	1	1
3)		6	3	3	1	0	1

Modules tested: V248, V3875, L3626, V147250, L143376.

Table 6.319: $L_2(61) < E_8$, $p = 0, 31$ or $p \nmid |H|$

		V_{248}			
		62_a	62_g	62_k	62_m
1)	P	1	1	1	1

Modules not used (elements of large order): V3875, V147250.

Modules tested: V248.

Table 6.320: $L_2(61) < E_8$, $p = 5$

		V_{248}
		62_a
1)	P	4

Modules not used (elements of large order): V3875, V147250.

Modules tested: V248.

Table 6.321: $L_2(61) < E_8$, $p = 3$

		V_{248}	
		62_a	62_c
1)	P	2	2

Modules not used (elements of large order): V3875, V147250, L113243.

Modules tested: V248.

Table 6.322: $L_2(61) < E_8$, $p = 2$

Not recomputed: no Brauer table available in GAP. See the Memoir.

Table 6.323: $L_3(3) < E_8$, $p = 0$ or $p \nmid |H|$

		V_{248}											
		1	12	13	16_a	16_b	16_c	16_d	26_a	26_b	26_c	27	39
1)	P	0	0	1	0	0	0	0	1	2	2	1	2
2)	P	0	2	0	0	0	0	0	4	0	0	3	1
3)		1	0	0	0	0	0	0	1	2	2	0	3
4)		1	0	2	0	0	0	0	3	2	2	0	1

	V_{248}											
	1	12	13	16_a	16_b	16_c	16_d	26_a	26_b	26_c	27	39
5)	2	0	1	1	1	1	1	0	1	1	0	3
6)	2	0	3	1	1	1	1	2	1	1	0	1
7)	3	0	4	2	2	2	2	1	0	0	0	1
8)	4	4	0	1	1	1	1	1	1	1	2	0
9)	5	4	1	2	2	2	2	0	0	0	2	0
10)	6	4	0	2	2	2	2	0	0	0	1	1
11)	8	0	0	0	0	0	0	1	1	1	6	0
12)	9	0	1	1	1	1	1	0	0	0	6	0
13)	10	0	0	1	1	1	1	0	0	0	5	1
14)	14	0	0	0	0	0	0	7	1	1	0	0
15)	15	0	1	1	1	1	1	6	0	0	0	0

Memoir rows not reproduced: 5) (not compatible with V3875).

Modules tested: V248, V3875, V147250.

Table 6.324: $L_3(3) < E_8$, $p = 13$

		V_{248}							
		1	11	13	16	26_a	26_b	26_c	39
1)	P	0	1	1	1	1	2	2	2
2)		1	0	0	0	1	2	2	3
3)		1	0	2	0	3	2	2	1
4)		1	1	2	5	0	1	1	2
5)		2	0	1	4	0	1	1	3
6)		2	0	3	4	2	1	1	1
7)	P, N	2	5	0	3	4	0	0	1
8)		3	0	4	8	1	0	0	1
9)	N	8	6	0	6	1	1	1	0
10)	N	9	6	1	10	0	0	0	0
11)	N	10	5	0	9	0	0	0	1
12)		14	0	0	0	7	1	1	0
13)		15	0	1	4	6	0	0	0

Modules tested: V248, V3875, V147250.

Table 6.325: $L_3(3) < E_8, p = 2$

		V_{248}						
		1	12	16_a	16_b	16_c	16_d	26
1)	P, N	4	3	0	0	0	0	8
2)	P, N	6	4	1	1	1	1	5
3)	N	8	5	2	2	2	2	2
4)	N	14	0	0	0	0	0	9
5)	N	16	1	1	1	1	1	6

Modules tested: V248, V3875, L3626, V147250, L143376.

Table 6.326: $L_3(5) < E_8, p \neq 2, 5$

		V_{248}	
		124_e	124_f
1)	P	1	1

Modules tested: V248, V3875, V147250.

Table 6.327: $L_3(5) < E_8, p = 2$

		V_{248}
		124_b
1)	P	2

Memoir rows not reproduced: 2) (not compatible with V3875; nor with L3626; nor with V147250; nor with L143376).

Modules tested: V248, V3875, L3626, V147250, L143376.

Table 6.328: $L_4(3) < E_8, p = 2$

		V_{248}		
		1	26_a	26_b
1)		14	8	1

Memoir rows not reproduced: 1) (not compatible with L3626); 2) (not compatible with L3626).

Modules tested: V248, V3875, L3626, V147250, L143376.

Table 6.329: $L_4(5) < E_8$, $p = 2$ (irreducible on V_{248})

Not recomputed: no Brauer table available in GAP. See the Memoir.

Table 6.330: $U_3(3) < E_8$, $p = 0$ or $p \nmid |H|$

	V_{248}													
	1	6	7_a	7_b	7_c	14	21_a	21_b	21_c	27	28_a	28_b	32_a	32_b
1)	1	0	0	0	0	3	1	0	0	0	1	1	2	2
2)	2	0	1	0	0	0	1	1	1	0	2	2	1	1
3)	2	0	1	1	1	2	1	0	0	0	2	2	1	1
4)	3	0	0	0	0	0	0	1	1	1	2	2	1	1
5)	3	0	1	2	2	1	1	2	2	1	0	0	1	1
6)	3	0	2	2	2	1	1	0	0	0	3	3	0	0
7)	4	0	4	0	0	1	4	0	0	2	0	0	1	1
8)	4	0	2	3	3	0	1	2	2	1	1	1	0	0
9)	5	0	5	1	1	0	4	0	0	2	1	1	0	0
10)	6	0	13	0	0	5	0	0	0	3	0	0	0	0
11)	8	0	0	0	0	1	0	0	0	6	0	0	1	1
12)	9	0	1	1	1	0	0	0	0	6	1	1	0	0
13)	17	14	0	2	2	7	1	0	0	0	0	0	0	0
14)	52	0	26	0	0	1	0	0	0	0	0	0	0	0

Modules tested: V248, V3875, V147250.

Table 6.331: $U_3(3) < E_8$, $p = 7$

		V_{248}											
		1	6	7_a	7_b	7_c	14	21_a	21_b	21_c	26	28_a	28_b
1)	P, N	1	2	0	0	0	0	3	1	1	5	0	0
2)	P, N	1	4	0	0	0	3	1	0	0	4	1	1
3)	N	2	2	1	0	0	0	1	1	1	2	2	2
4)	N	2	2	1	1	1	2	1	0	0	2	2	2
5)		3	0	2	2	2	1	1	0	0	0	3	3
6)	N	4	2	0	0	0	0	0	1	1	3	2	2
7)	N	4	2	1	2	2	1	1	2	2	3	0	0
8)		5	0	2	3	3	0	1	2	2	1	1	1
9)	N	6	2	4	0	0	1	4	0	0	4	0	0
10)	N	7	0	5	1	1	0	4	0	0	2	1	1
11)	N	9	0	13	0	0	5	0	0	0	3	0	0
12)	N	14	2	0	0	0	1	0	0	0	8	0	0

		V_{248}											
		1	6	7_a	7_b	7_c	14	21_a	21_b	21_c	26	28_a	28_b
13)	N	15	0	1	1	1	0	0	0	0	6	1	1
14)		17	14	0	2	2	7	1	0	0	0	0	0
15)		52	0	26	0	0	1	0	0	0	0	0	0

Modules tested: V248, V3875, V147250, L147002.

Table 6.332: $U_3(8) < E_8$, $p \neq 2, 3$

		V_{248}				
		1	56	57_a	57_b	133_a
1)		1	0	1	1	1
2)		3	2	0	0	1

Modules tested: V248, V3875, V147250.

Table 6.333: $U_3(8) < E_8$, $p = 3$

		V_{248}		
		1	56	133_a
1)	N	3	2	1

Modules tested: V248, V3875, V147250, L113243.

Table 6.334: $U_4(2) < E_8$, $p = 0$ or $p \nmid |H|$

		V_{248}													
		1	5_a	5_b	6	10_a	10_b	15_a	20	24	40_a	40_b	45_a	45_b	64
1)	P	0	0	0	0	0	0	2	0	0	0	0	1	1	2
2)	P	0	1	1	0	1	1	0	0	2	1	1	1	1	0
3)		11	0	0	12	2	2	7	1	0	0	0	0	0	0
4)		24	10	10	0	5	5	0	0	1	0	0	0	0	0

Modules tested: V248, V3875, V147250.

Table 6.335: $U_4(2) < E_8$, $p = 5$

		V_{248}													
		1	5_a	5_b	6	10_a	10_b	15_a	20	23	40_a	40_b	45_a	45_b	58
1)	P	0	0	0	2	0	0	2	0	0	0	0	1	1	2
2)	N	2	1	1	0	1	1	0	0	2	1	1	1	1	0
3)		11	0	0	12	2	2	7	1	0	0	0	0	0	0
4)		25	10	10	0	5	5	0	0	1	0	0	0	0	0

Modules tested: V248, V3875, V147250.

Table 6.336: $PSp_4(5) < E_8$, $p = 2$

		V_{248}					
		1	12_a	12_b	40	64	104_b
1)	P	0	0	0	1	0	2
2)	P, N	8	4	4	2	1	0

Modules tested: V248, V3875, L3626, V147250, L143376.

Table 6.337: $Sp_6(2) < E_8$, $p = 0, 5$ or $p \nmid |H|$

		V_{248}				
		1	7	21_a	27	35_a
1)		4	6	5	1	2

Modules tested: V248, V3875, V147250.

Table 6.338: $Sp_6(2) < E_8$, $p = 7$

		V_{248}				
		1	7	21_a	26	35_a
1)		5	6	5	1	2

Modules tested: V248, V3875, V147250, L147002.

Table 6.339: $Sp_6(2) < E_8$, $p = 3$

	V_{248}				
	1	7	21	27	35
1)	4	6	5	1	2

Modules tested: V248, V3875, V147250, L113243.

Table 6.340: $\Omega_8^+(2) < E_8$, $p \neq 2$

	V_{248}				
	1	28	35_a	35_b	35_c
1)	3	5	1	1	1

Modules tested: V248, V3875, V147250.

Table 6.341: $G_2(3) < E_8$, $p \neq 2, 3$

	V_{248}					
	1	14	64_a	64_b	78	91_a
1)	1	0	0	0	2	1
2)	1	2	1	1	0	1

Modules tested: V248, V3875, V147250.

Table 6.342: $G_2(3) < E_8$, $p = 2$

	V_{248}					
	1	14	64_a	64_b	78	90_a
1)	2	0	0	0	2	1
2)	2	2	1	1	0	1

Modules tested: V248, V3875, L3626, V147250, L143376.

Table 6.343: ${}^3D_4(2) < E_8, p \neq 2, 3$

	V_{248}		
	1	26	52
1)	14	7	1

Memoir rows not reproduced: 1) (not feasible on the Memoir's modules (all element orders, full power maps)).
Modules tested: V248, V3875, V147250.

Table 6.344: ${}^3D_4(2) < E_8, p = 3$

		V_{248}			
		1	25	52	196
1)	P	0	0	1	1
2)	N	21	7	1	0

Modules tested: V248, V3875, V147250, L113243.

Table 6.345: ${}^2F_4(2)' < E_8, p \neq 2, 3, 5$

	V_{248}			
	1	27_a	27_b	78
1)	8	3	3	1

Modules tested: V248, V3875, V147250.

Table 6.346: ${}^2F_4(2)' < E_8, p = 5$

		V_{248}			
		1	27_a	27_b	78
1)	N	8	3	3	1

Modules tested: V248, V3875, V147250.

Table 6.347: ${}^2F_4(2)' < E_8, p = 3$

	V_{248}					
	1	27_a	27_b	77	124_a	124_b
1)	0	0	0	0	1	1
2)	9	3	3	1	0	0

Modules tested: V248, V3875, V147250, L113243.

Table 6.348: ${}^2B_2(8) < E_8, p = 0$ or $p \nmid |H|$

		V_{248}					
		1	14_a	14_b	64	65_a	91
1)	P	0	1	1	1	1	1
2)		1	1	1	2	0	1

Memoir rows not reproduced: 2) (not compatible with V3875; nor with V147250); 4) (not compatible with V3875; nor with V147250); 5) (not compatible with V3875; nor with V147250).

Modules tested: V248, V3875, V147250.

Table 6.349: ${}^2B_2(8) < E_8, p = 13$

	V_{248}					
	1	14_a	14_b	35	65_a	91
1) P	1	2	2	1	1	1
2) P	3	3	3	2	0	1

Memoir rows not reproduced: 1) (not compatible with V3875; nor with V147250); 3) (not compatible with V3875; nor with V147250); 4) (not compatible with V3875; nor with V147250).

Modules tested: V248, V3875, V147250.

Table 6.350: ${}^2B_2(8) < E_8, p = 7$

		V_{248}				
		1	14_a	14_b	64	91
1)	P	1	1	1	2	1
2)		8	4	4	2	0

Memoir rows not reproduced: 2) (not compatible with V3875; nor with V147250; nor with L147002); 3) (not compatible with V3875; nor with V147250; nor with L147002); 5) (not compatible with V3875; nor with V147250; nor with L147002).

Modules tested: V248, V3875, V147250, L147002.

Table 6.351: ${}^2B_2(8) < E_8, p = 5$

		V_{248}							
		1	14_a	14_b	35_a	35_b	35_c	63	65_a
1)	P	1	2	2	0	0	0	2	1
2)		3	0	0	4	0	3	0	0
3)		3	2	2	0	0	0	3	0
4)		3	5	5	1	1	1	0	0

Modules tested: V248, V3875, V147250.

Table 6.352: ${}^2B_2(32) < E_8, p = 5$

		V_{248}	
		124_a	124_b
1)	P	1	1

Modules not used (elements of large order): V3875, V147250.

Modules tested: V248.